CIP-B106: Mobile and IoT Forensics

Case Study 3:Echo Show 8 IoT Evidence Investigation

Student Name: Favour Ifeyinwa Madu

Student ID: [C11-26-DFIT17314**]**

Date: [11-18 October, 2026]

Platform Used

Kali Linux (VMware Workstation)

C11-26-DFIT17314_CIP-B106_18102026

CIP-B106_ECHO_SHOW8_IOT_EVIDENCE_26-DFIT17314_FAVOUR_MADU_18102026

INSTRUCTOR: AMINU IDRIS, AMCPN | FOUNDER, ICDFA

Forensic Examination Report – Case File CIPB106 EchoShow8

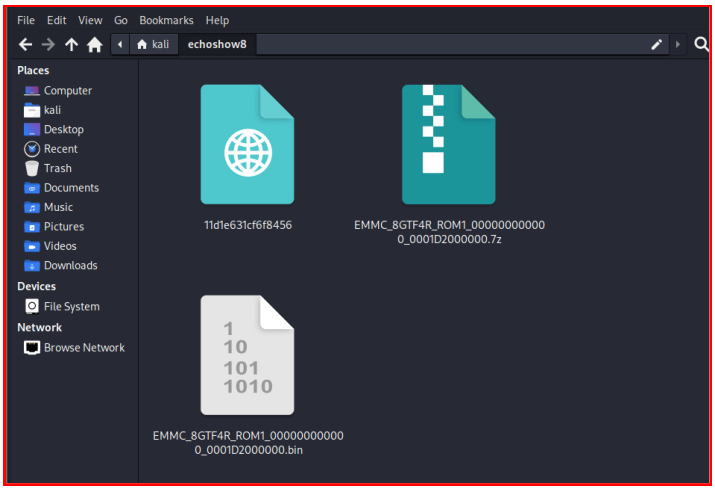
Report status: This report is based on the examiner's working notes in Annex A. Any value shown in square brackets, for example [insert], appears only in a screenshot or tool output. It must be copied exactly from that source before the report is finalized. No value has been estimated or assumed.

1. Case Question and Approach

Case question: What can a careful examination of the Echo Show 8 eMMC and user data evidence establish about device identity, operating system and configuration, user-related data, network and web activity, phone communication, multimedia evidence, and the sequence of events during a justified examination window?

This is a device-level reconstruction of a training dataset. It does not involve cloud acquisition, account access, or contact with a live Echo device, and it does not decide whether any person committed an offence. The report uses neutral terms such as device, device user, account, contact, and person of interest only where the evidence supports them. Each finding is classified as one of the following: **Verified fact** (directly observed in a named artefact and reproducible); **Reasonable inference** (an interpretation supported by one or more verified facts); **Limitation** (a constraint on what the evidence can show); or **Unknown** (not established by the available material).






```
┌──(kali㉿kali)-[~/echoshow8]
└─$ 7z x EMMC_8GTF4R_ROM1_000000000000_0001D2000000.7z

7-Zip 26.01 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-04-27
 64-bit locale=en_US.UTF-8 Threads:128 OPEN_MAX:4096, ASM

Scanning the drive for archives:
1 file, 2396000768 bytes (2286 MiB)

Extracting archive: EMMC_8GTF4R_ROM1_000000000000_0001D2000000.7z
--
Path = EMMC_8GTF4R_ROM1_000000000000_0001D2000000.7z
Type = 7z
Physical Size = 2396000768
Headers Size = 202
Method = LZMA2:24
Solid = -
Blocks = 1

Everything is Ok

Size:       7818182656
Compressed: 2396000768
```

UNZIP the file and verify the image file hash value

```
┌──(kali㉿kali)-[~/echoshow8]
└─$ ls -l
total 9974808
-rw-rw-r-- 1 kali kali       5877 Jul  6 23:16 11d1e631cf6f8456
-rw-rw-r-- 1 kali kali 2396000768 Oct 11 09:18 EMMC_8GTF4R_ROM1_000000000000_0001D2000000.7z
-rw-rw-r-- 1 kali kali 7818182656 Dec 29  2023 EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin

┌──(kali㉿kali)-[~/echoshow8]
└─$ ll

┌──(kali㉿kali)-[~/echoshow8]
└─$ ll EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
-rw-rw-r-- 1 kali kali 7818182656 Dec 29  2023 EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin

┌──(kali㉿kali)-[~/echoshow8]
└─$ md5sum EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
b58fa04967161c158723c7b00a636533  EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
```

Hash verification matches the one instruction file provided;
b58fa04967161c158723c7b00a636533

Setting up loop device

sudo losetup --partscan --find --show --read-only EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin

```
┌──(kali㉿kali)-[~/echoshow8]
└─$ sudo losetup --partscan --find --show --read-only EMMC_8GTF4R_ROM1_000000000000_0001D2000000.b
in
/dev/loop0

┌──(kali㉿kali)-[~/echoshow8]
└─$ losetup -a
/dev/loop0: []: (/home/kali/echoshow8/EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin)
```

Note

- **sudo:** root privilege
- **losetup:** Command to set up loop devices.
- **--partscan:** This option tells losetup to scan partitions on the device.
- **--find:** This option instructs losetup to find the first available loop device automatically.
- **--show:** This option makes losetup display the name of the loop device it finds.
- **--read-only:** This option sets up the loop device as read-only.

CHECK PARTITION NUMBER OF BLOCKED DEVICE

```
┌──(kali㉿kali)-[~/echoshow8]
└─$ lsblk /dev/loop0
NAME      MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0    0  7.3G  1 loop
├─loop0p1 259:0    0    1M  1 part
├─loop0p2 259:1    0    1M  1 part
├─loop0p3 259:2    0    5M  1 part
├─loop0p4 259:3    0    9M  1 part
├─loop0p5 259:4    0    5M  1 part
├─loop0p6 259:5    0  512K  1 part
├─loop0p7 259:6    0   40M  1 part
├─loop0p8 259:7    0   32M  1 part
├─loop0p9 259:8    0 39.5M  1 part
├─loop0p10 259:9   0    1M  1 part
├─loop0p11 259:10  0   10M  1 part
├─loop0p12 259:11  0  286M  1 part
├─loop0p13 259:12  0    1M  1 part
├─loop0p14 259:13  0    1M  1 part
├─loop0p15 259:14  0   15M  1 part
├─loop0p16 259:15  0   32M  1 part
├─loop0p17 259:16  0   15M  1 part
├─loop0p18 259:17  0    1M  1 part
├─loop0p19 259:18  0    1M  1 part
├─loop0p20 259:19  0   16M  1 part
├─loop0p21 259:20  0    3G  1 part
├─loop0p22 259:21  0  256M  1 part
└─loop0p23 259:22  0  3.5G  1 part
```

CHECK PARTITION LAYOUT OF IMAGE

```
┌──(kali㉿kali)-[~/echoshow8]
└─$ mmls EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

     Slot      Start        End          Length       Description
000:  Meta     0000000000   0000000000   0000000001   Safety Table
001:  ------   0000000000   0000000063   0000000064   Unallocated
002:  Meta     0000000001   0000000001   0000000001   GPT Header
003:  Meta     0000000002   0000000033   0000000032   Partition Table
004:  000      0000000064   0000002111   0000002048   kb
005:  001      0000002112   0000004159   0000002048   dkb
006:  002      0000004160   0000014399   0000010240   tee1
007:  003      0000014400   0000032767   0000018368   tee2
008:  004      0000032768   0000043007   0000010240   persist
009:  005      0000043008   0000044031   0000001024   misc
010:  006      0000044032   0000125951   0000081920   recovery
011:  007      0000125952   0000191487   0000065536   swdl
012:  008      0000191488   0000272383   0000080896   metadata
013:  009      0000272384   0000274431   0000002048   boot_para
014:  010      0000274432   0000294911   0000020480   nvcfg
015:  011      0000294912   0000880639   0000585728   vendor
016:  012      0000880640   0000882687   0000002048   spmfw
017:  013      0000882688   0000884735   0000002048   sspm_1
018:  014      0000884736   0000915455   0000030720   cam_vpu1
019:  015      0000915456   0000980991   0000065536   cam_vpu2
020:  016      0000980992   0001011711   0000030720   cam_vpu3
021:  017      0001011712   0001013759   0000002048   lk
022:  018      0001013760   0001015807   0000002048   lk2
023:  019      0001015808   0001048575   0000032768   boot
024:  020      0001048576   0007405567   0006356992   system
025:  021      0007405568   0007929855   0000524288   cache
026:  022      0007929856   0015269854   0007339999   userdata
027:  ------   0015269855   0015269887   0000000033   Unallocated
```

(**mmls output**)	(**lsblk output**)	(**Partition Name**)
020	p21	System Partition
022	p23	Userdata Partition

Mount SYSTEM PARTITION P21

```
┌──(kali㉿kali)-[~/echoshow8]
└─$ sudo mkdir /mnt/echoshow8
[sudo] password for kali:
```

```
┌──(kali㉿kali)-[~/echoshow8]
└─$ sudo mount /dev/loop0p21 /mnt/echoshow8
mount: /mnt/echoshow8: WARNING: source write-protected, mounted read-only.
```

```
┌──(kali㉿kali)-[~/echoshow8]
└─$ cd /mnt/echoshow8

┌──(kali㉿kali)-[/mnt/echoshow8]
└─$ ls
acct        data           init.environ.rc         init.recovery.mt8183.rc  odm       storage
bin         default.prop   init.fosflags.sh        init.usb.configfs.rc     oem       sys
bugreports  dev            init.inlocservice.rc    init.usb.rc              persist   system
cache       etc            init.meshmgrservice.rc  init.whad_cc.rc          proc      ueventd.rc
charger     init           init.perfmonitord.rc    init.zygote32.rc         product   vendor
config      init.aosp.rc   init.perfrecoveryd.rc   lost+found               sbin      verity_key
d           init.build.rc  init.rc                 mnt                      sdcard
```

MOUNT USER DATA PARTITION P23

```
┌──(kali㉿kali)-[/mnt/echoshow8]
└─$ sudo mkdir /mnt/echoshow8-2

┌──(kali㉿kali)-[/mnt/echoshow8]
└─$ sudo mount /dev/loop0p23 /mnt/echoshow8-2
mount: /mnt/echoshow8-2: cannot mount /dev/loop0p23 read-only.
       dmesg(1) may have more information after failed mount system call.

┌──(kali㉿kali)-[/mnt/echoshow8]
└─$ sudo dmesg | tail -n 1
[293132.349231] EXT4-fs (loop0p23): write access unavailable, cannot proceed (try mounting with no
load)
```

- The ***noload*** option refers to a mount option that prevents the kernel from loading the journal of the file system.
- The journal (in file systems such as ext3 and ext4) keeps track of changes to the file system for faster recovery in case of a system crash or unexpected shutdown.
- Using ***noload*** is necessary when dealing with forensic analysis where you want to examine the file system without making any modifications.

```
└─$ sudo mount -o ro,noload /dev/loop0p23 /mnt/echoshow8-2

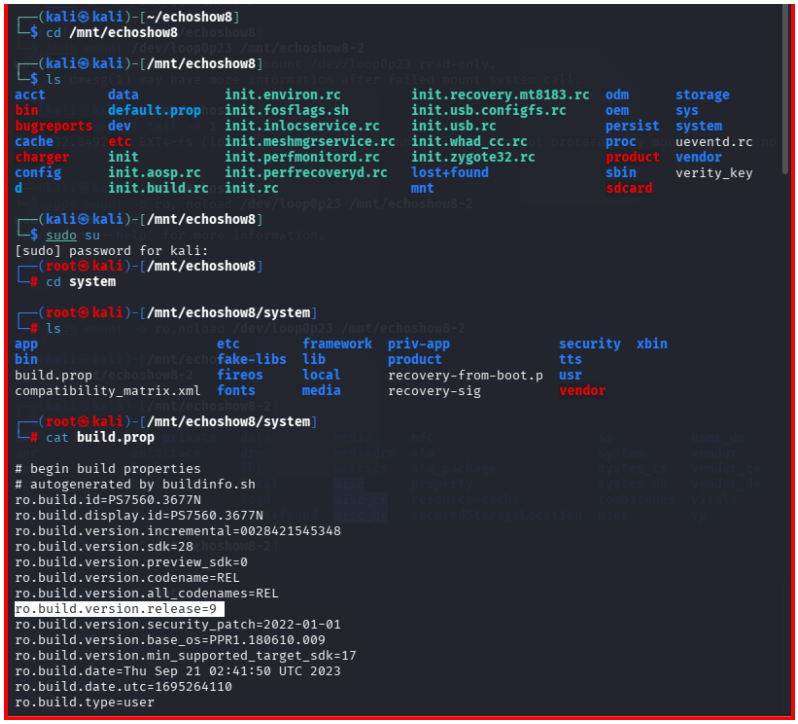
┌──(kali㉿kali)-[/mnt/echoshow8]
└─$ cd /mnt/echoshow8-2

┌──(kali㉿kali)-[/mnt/echoshow8-2]
└─$ ls
adb           app-private   data        media     nfc                    ss          user_de
anr           autotrace     drm         mediadrm  ota                    system      vendor
app           backup        fht         metrics   ota_package            system_ce   vendor_ce
app-asec      bootchart     local       misc      property               system_de   vendor_de
app-ephemeral cache         logd        misc_ce   resource-cache         tombstones  vitals
app-lib       dalvik-cache  lost+found  misc_de   securedStorageLocation  user        vp
```

Linux Commands Used in This Lab

Command	Purpose	Command Syntax Used	Explanation
mmls	To display the partition layout of a disk image.	mmls disk_image.img	Display partition layout of disk image named disk_image.img
losetup	To set up loop devices. Loop devices are often used to mount filesystems stored in regular files for testing or recovery purposes	losetup -fP /path/to/disk_image.img	-f => automatically find the first available loop device -P => automatically create and associate the loop device with all partitions within the specified file
lsblk	To list information about block devices.	lsblk /dev/loop0	Display all information about loop device at loop0 including partitions
mount	To attach a filesystem (a collection of files and directories) to a specific directory	mount /dev/loop0p21 /mnt/mydisk	Mount source (loop device at loop0p21) to mount point (/mnt/mydisk)

Display Partitions of the eMMC Image



OS Info Found in ‘build.prop’ File

```
# maximum backup size for factory reset whitelist in kB
ro.recovery.wl.maxstore=921600

# Reserve WHA server ports.
sys.sysctl.ip_loc_res_ports=55442-55445

# Allow boosting of CPUs up to fifteen seconds.
persist.perfboost.time=15000

# Extended smarthome discovery timeout in msec
persist.sh_discovery_timeout=40000

#enable OTA download via metered network
persist.sys.ota_allow_metered=true
#
# ADDITIONAL_BUILD_PROPERTIES
#
ro.treble.enabled=true
persist.sys.dalvik.vm.lib.2=libart.so
dalvik.vm.isa.arm.variant=cortex-a53
dalvik.vm.isa.arm.features=default
net.bt.name=Android
dalvik.vm.stack-trace-dir=/data/anr
ro.build.lab126.buildtype=Nightly
ro.build.lab126.project=athena_fireos_ship_7560
ro.build.lab126.build=3677
ro.build.version.serial=3677
ro.build.configuration=knightfamily
ro.build.version.fireos=7.0
ro.build.version.number=0028421545348
ro.build.mktg.fireos=Fire OS 7.5.6.0
ro.build.version.name=Fire OS 7.5.6.0 (PS7560/3677)
ro.product.package_name=com.amazon.athena.android.os
ro.expect.recovery_id=0×eb4b1c0196bad041b21785b5f1087dedb9c2d73c0000000000000000000000000
```

Fire OS Vs Android

Fire OS is a customized version of Android developed by Amazon for its Fire devices, like the Echo Show series.

The Fire OS is Android-based, but it has its own app store and ecosystem.

Amazon customizes and pre-installs certain apps on its Fire devices.

Installed System Apps on the Device

```
┌──(root㉿kali)-[/mnt/echoshow8/system]
└─# ls
app                  etc        framework  priv-app                 security  xbin
bin                  fake-libs  lib        product                  tts
build.prop           fireos     local      recovery-from-boot.p     usr
compatibility_matrix.xml  fonts  media      recovery-sig             vendor

┌──(root㉿kali)-[/mnt/echoshow8/system]
└─# cd app

┌──(root㉿kali)-[/mnt/echoshow8/system/app]
└─# ls
AmazonDeviceMetrics          Email                       PrintRecommendationService
AmazonWebView                ExtShared                   Protips
AmazonWirelessMetricsService  fdrw                       QuickSearchBox
Bluetooth                    HTMLViewer                  SecureElement
BluetoothMidiService         KeyChain                    SidewalkDeviceService
BookmarkProvider             KnightBluetoothAVRCPClient  SimAppDialog
BuiltInPrintService          KnightCaptivePortalLogin    Traceur
Calendar                     LiveWallpapersPicker        WallpaperBackup
CertInstaller                Music                       WallpaperPicker
CompanionDeviceManager       NfcNci                      WAPPushManager
CtsShimPrebuilt              PacProcessor                whad_app
DeskClock                    PicoTts
```

DEVICE SERIAL NUMBER

```
┌──(root㉿kali)-[/mnt/echoshow8-2]
└─# cd data/com.android.settings/shared_prefs

┌──(root㉿kali)-[/mnt/echoshow8-2/data/com.android.settings/shared_prefs]
└─# ls
account-verification.xml                    com.android.settings.FLAGS.xml
alexa-pairing-mode.xml                      com.android.settings_preferences.xml
bluetooth_settings.xml                      com.android.settings.xml
com.amazon.settings.bluetooth_prefs_key.xml development.xml
com.amazon.settings.CONFIG_BUNDLE_STORE.xml device_disconnection_times.xml
com.amazon.settings.DNSVR_STORE.xml         deviceNameSharedPref.xml
com.amazon.settings.SETTINGS_STORE.xml      KBluetoothControllerSharedPrefs.xml
com.amazon.settings.wifi.xml                KnightSettingsMetrics.xml

┌──(root㉿kali)-[/mnt/echoshow8-2/data/com.android.settings/shared_prefs]
└─# cat deviceNameSharedPref.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <string name="deviceNameKey">Amazing Assistant</string>
</map>

┌──(root㉿kali)-[/mnt/echoshow8-2/data/com.android.settings/shared_prefs]
└─# cat deviceNameSharedPref.xml

┌──(root㉿kali)-[/mnt/echoshow8-2/data/com.android.settings/shared_prefs]
└─# cat com.amazon.settings.DNSVR_STORE.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <set name="devicepreferenceslist">
        <string>G6G1GG10141403SD|A0565892JJWBS2358X2D</string>
        <string>ead6f3de7cf46a2be0aabde8de37b7a|A0609322RBQIQJZ3HIL3</string>
        <string>1D9A9AC3FFF5419EBA1BAAE12C1FEF3A|A01916772KYQTFFKFZ3M6</string>
    </set>
</map>
```

DEVICE NAME AND SERIAL NUMBER

```
┌──(root㉿kali)-[/mnt/echoshow8-2]
└─# cd data/amazon.speech.sim/files

┌──(root㉿kali)-[/mnt/echoshow8-2/data/amazon.speech.sim/files]
└─# ls
aedModels  device_capabilities_api.amazonalexa.com  models  nttModels  tmpModels

┌──(root㉿kali)-[/mnt/echoshow8-2/data/amazon.speech.sim/files]
└─# cat device_capabilities_api.amazonalexa.com
{"endpoints":[{"endpointId":"A15996VY63BQ2D@G6G1GG10141403SD","legacyIdentifier":{"deviceType":"A1
5996VY63BQ2D","deviceSerialNumber":"G6G1GG10141403SD"},"manufacturerName":"amazon","description":"
Amazon Alexa","friendlyName":"Amazing Assistant","displayCategoriesMetadata":[{"name":"ALEXA_VOICE
_ENABLED","providerType":"RESERVED_INTERNAL_ONLY_DYNAMIC","owningPackages":["amazon.speech.sim"],"
clientRef":"amazon.speech.sim/amazon.speech.simclient.display.devicecapability.WindowCapabilityMan
ager"},{"name":"CAMERA","providerType":"RESERVED_INTERNAL_ONLY_DYNAMIC","owningPackages":["com.ama
zon.mlv"],"clientRef":"com.amazon.mlv/com.amazon.mlvlib.discovery.DiscoveryManager"}],"capabilitie
s":[{"type":"AlexaInterface","interface":"Alexa.Display.Window","version":"1.1","owningPackages":[
"amazon.speech.sim"],"clientRef":"amazon.speech.sim/amazon.speech.simclient.display.devicecapabili
ty.WindowCapabilityManager","providerType":"RESERVED_INTERNAL_ONLY_DYNAMIC","configurations":{"tem
plates":[{"id":"MMFullScreen","type":"STANDARD","configuration":{"interactionModes":["MMInteractio
nMode"],"sizes":[{"value":{"value":{"width":1280,"height":800},"unit":"PIXEL"},"id":"MultimodalFul
lScreenSize","type":"DISCRETE"}]}},{"id":"MMPartialScreen","type":"OVERLAY","configuration":{"inte
ractionModes":["MMInteractionMode"],"sizes":[{"value":{"value":{"width":426,"height":800},"unit":"
```

Profile ID, User's Name and Encryption Key

LIST OF USERS

```
┌──(kali㉿kali)-[/mnt/echoshow8]
└─$ cd /mnt/echoshow8-2

┌──(kali㉿kali)-[/mnt/echoshow8-2]
└─$ cd system/users

┌──(kali㉿kali)-[/mnt/echoshow8-2/system/users]
└─$ ls
0  0.xml  sorteduserlist.xml  userlist.xml
```

File	Description
0.xml	• Information about the primary user • It contains elements such as name, icon, flags, serial number, etc. that identify the primary user and its settings
userlist.xml	• Information about users created on the device like User ID
sorteduserlist.xml	• List of user IDs

```
┌──(kali㉿kali)-[/mnt/echoshow8-2/system/users]
└─$ cat userlist.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<users nextSerialNumber="10" version="7">
    <guestRestrictions>
        <restrictions no_sms="true" no_install_unknown_sources="true" no_config_wifi="true" no
_outgoing_calls="true" />
    </guestRestrictions>
    <deviceOwnerUserId id="-10000" />
    <user id="0" />
</users>

┌──(kali㉿kali)-[/mnt/echoshow8-2/system/users]
└─$ cat sorteduserlist.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<users>
<user id="0" />
</users>
```

NAME OF USER (PRIMARY USER WITH ID 0

```
┌──(kali㉿kali)-[/mnt/echoshow8-2/system/users]
└─$ cat 0.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<user id="0" serialNumber="0" flags="19" created="0" lastLoggedIn="1698016596033" lastLoggedIn
Fingerprint="Amazon/athena/athena:9/PS7560.3677N/0028421545216:user/amz-p,release-keys" icon="
/data/system/users/0/photo.png" profileBadge="0">
    <name>Sarfraz</name>
    <restrictions />
    <ignorePrepareStorageErrors>true</ignorePrepareStorageErrors>
</user>
┌──(kali㉿kali)-[/mnt/echoshow8-2/system/users]
└─$ date -u -d @

┌──(kali㉿kali)-[/mnt/echoshow8-2/system/users]
└─$ date -u -d @16980165960
Mon Jan 30 04:46:00 PM UTC 2508

┌──(kali㉿kali)-[/mnt/echoshow8-2/system/users]
└─$ date -u -d @1698016596
Sun Oct 22 11:16:36 PM UTC 2023
```

USER ACTIVITY STATISTICS

```
┌──(kali㉿kali)-[/mnt/echoshow8-2/system]
└─$ cd usagestats/0

┌──(kali㉿kali)-[/mnt/echoshow8-2/system/usagestats/0]
└─$ ls
daily  monthly  version  weekly  yearly

┌──(kali㉿kali)-[/mnt/echoshow8-2/system/usagestats/0]
└─$ cd daily

┌──(kali㉿kali)-[/mnt/…/system/usagestats/0/daily]
└─$ ls
1697329601115  1697502596344  1697675612477  1697848962840  1698022332424
1697416099195  1697589171089  1697762397254  1697935512409

┌──(kali㉿kali)-[/mnt/…/system/usagestats/0/daily]
└─$ date -u -d @1697935512
Sun Oct 22 12:45:12 AM UTC 2023
```

DISPLAY CONTENTS OF FILE IN DAILY DIRECTORY

```
┌──(kali㉿kali)-[/mnt/…/system/usagestats/0/daily]
└─$ cat 1697935512409 | more
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<usagestats version="1" endTime="85500007">
    <interactive count="11" time="1697860050653" />
    <non-interactive count="11" time="1697859985758" />
    <keyguard-shown count="0" time="0" />
    <keyguard-hidden count="0" time="0" />
    <packages>
        <package lastTimeActive="-1697775240571" package="com.android.providers.calendar" time
Active="0" lastEvent="0" />
        <package lastTimeActive="78119740" package="com.amazon.device.rdmapplication" timeActi
ve="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.whisperjoin.middleware" t
imeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.alexa.beaconbroadcaster"
timeActive="0" lastEvent="0" />
        <package lastTimeActive="84396503" package="com.amazon.alexa.datastore.app" timeActive
="0" lastEvent="0" />
```

```
    </packages>
    <configurations />
    <event-log>
        <event time="3205245" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="3295445" package="com.amazon.android.service.wifiprofilemanager" flags="0
" type="12" />
        <event time="7784509" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="10286912" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="10485765" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="12863303" package="com.amazon.device.rdmapplication" flags="0" type="12"
/>
        <event time="12864014" package="com.amazon.device.rdmapplication" flags="0" type="12"
/>
        <event time="14374522" package="com.amazon.android.service.wifiprofilemanager" flags="
0" type="12" />
        <event time="17728673" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="17728917" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="18584708" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="23188181" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="24990376" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="26673439" package="android" flags="0" type="15" />
        <event time="26674042" package="com.amazon.paladin" class="com.amazon.paladin.app.MMLa
uncherActivity" flags="0" type="1" />
        <event time="27273501" package="android" flags="0" type="16" />
        <event time="27274016" package="com.amazon.paladin" class="com.amazon.paladin.app.MMLa
uncherActivity" flags="0" type="2" />
        <event time="32012083" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="32753281" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="33058222" package="android" flags="0" type="15" />
        <event time="33058267" package="com.amazon.paladin" class="com.amazon.paladin.app.MMLa
uncherActivity" flags="0" type="1" />
        <event time="33665721" package="android" flags="0" type="16" />
        <event time="33666168" package="com.amazon.paladin" class="com.amazon.paladin.app.MMLa
uncherActivity" flags="0" type="2" />
        <event time="38753315" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="39453897" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="39454065" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="46895540" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="46895735" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="54277309" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
--More--
```

User Account Information – system_ce

```
┌──(kali㉿kali)-[/mnt/echoshow8-2]
└─$ cd system_ce/0

┌──(kali㉿kali)-[/mnt/echoshow8-2/system_ce/0]
└─$ ls
accounts_ce.db      accounts_ce.db-wal  shortcut_service
accounts_ce.db-shm  recent_tasks        snapshots
```

```
┌──(kali㉿kali)-[/mnt/echoshow8-2/system_ce/0]
└─$ sqlite3 accounts_ce.db
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> .tables
accounts          android_metadata  authtokens        extras
sqlite> SELECT *FROM accounts;
1|AmazonSyncAccount-cba1f8294651551b7d9d16bef0970fdd79fca92f414c3338137185f6acf40f71|amazon.ac
count|
3|Amazon Video Sync|amazon.account|
4|AmazonSyncAccount-6957b355b974dde07bb47bf8cd86218da71f3ce800784526ba4345cda0abf63b|amazon.ac
count|
5|Sarfraz|com.amazon.account|
sqlite>
```

User Account Information – system_de

```
┌──(kali㉿kali)-[/mnt/echoshow8-2]
└─$ cd system_de/0

┌──(kali㉿kali)-[/mnt/echoshow8-2/system_de/0]
└─$ ls
accounts_de.db  accounts_de.db-journal  persisted_taskIds.txt  ringtones
```

```
┌──(kali㉿kali)-[/mnt/echoshow8-2/system_de/0]
└─$ sqlite3 accounts_de.db
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> .tables
accounts          debug_table       meta              visibility
android_metadata  grants            shared_accounts
sqlite> SELECT * FROM accounts;
1|AmazonSyncAccount-cba1f8294651551b7d9d16bef0970fdd79fca92f414c3338137185f6acf40f71|amazon.ac
count||1613617878925
3|Amazon Video Sync|amazon.account||1696280426425
4|AmazonSyncAccount-6957b355b974dde07bb47bf8cd86218da71f3ce800784526ba4345cda0abf63b|amazon.ac
count||1696280432902
5|Sarfraz|com.amazon.account||1696883981793
sqlite> SELECT * FROM meta;
auth_uid_for_type:com.amazon.account|10111
auth_uid_for_type:com.android.email|10126
auth_uid_for_type:com.android.exchange|10126
auth_uid_for_type:amazon.account|10118
sqlite> SELECT * FROM debug_table;
1|action_account_add|2021-02-18 03:11:18|10118|accounts|0
2|action_account_add|2023-10-02 21:00:18|10111|accounts|1
3|action_account_add|2023-10-02 21:00:26|10029|accounts|2
4|action_account_add|2023-10-02 21:00:32|10118|accounts|3
2|action_called_account_remove|2023-10-09 16:34:24|10111|accounts|4
2|action_account_remove|2023-10-09 16:34:24|10111|accounts|5
5|action_account_add|2023-10-09 16:39:41|10111|accounts|6
sqlite>
```

Data about Recently Used Apps

- **The /system_ce/0/recent_tasks contains the details of recently used apps**
- **The /system_ce/0/snapshots contains the screenshots of the recently used apps**
- **Recent tasks are stored in XML format**
- **The snapshots are stored in JPEG format.**
- **These have a file name that consists of a task ID followed by extension.**
- **For example, a snapshot file named 123.jpg means that it is the 123rd snapshot of the app.**
- **The snapshots are updated whenever an app is launched or resumed on the device.**

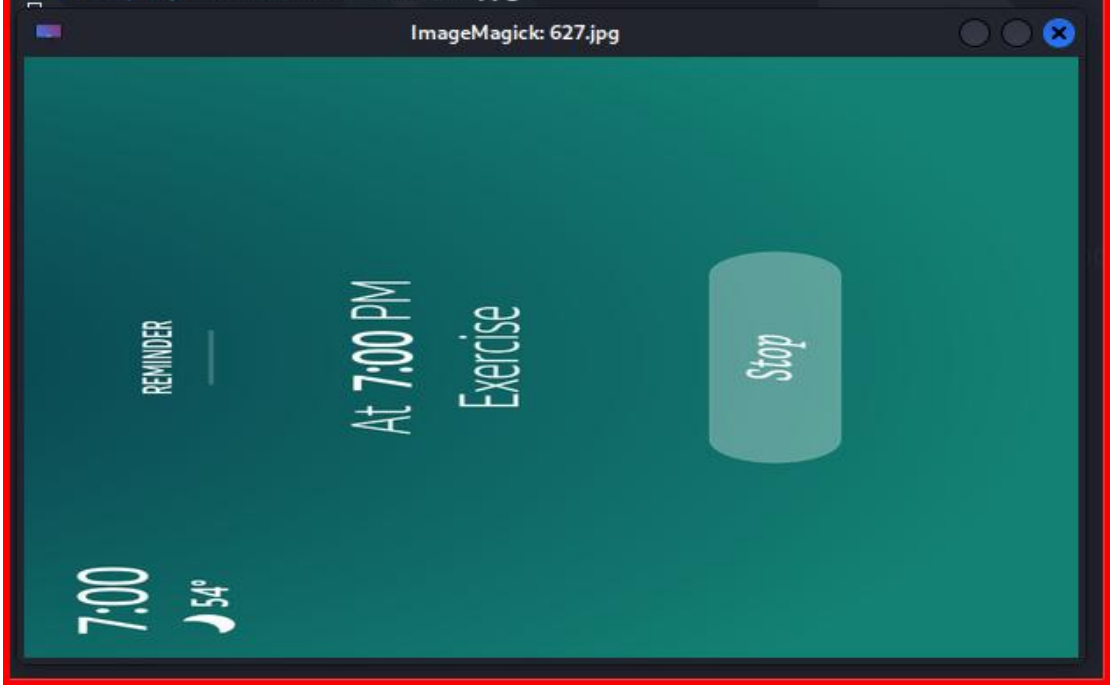
```
┌──(kali㉿kali)-[/mnt/echoshow8-2]
└─$ cd system_ce/0

┌──(kali㉿kali)-[/mnt/echoshow8-2/system_ce/0]
└─$ ls
accounts_ce.db      accounts_ce.db-wal  shortcut_service
accounts_ce.db-shm  recent_tasks        snapshots

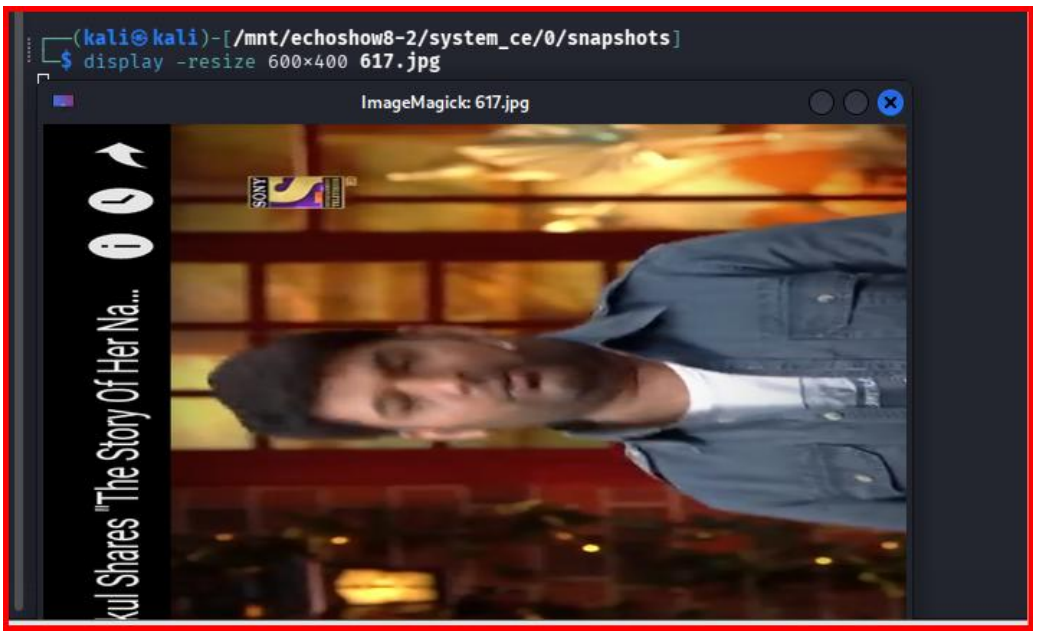
┌──(kali㉿kali)-[/mnt/echoshow8-2/system_ce/0]
└─$ cd snapshots

┌──(kali㉿kali)-[/mnt/echoshow8-2/system_ce/0/snapshots]
└─$ ls
617.jpg          623.jpg          624.jpg          625.jpg          627.jpg
617.proto        623.proto        624.proto        625.proto        627.proto
617_reduced.jpg  623_reduced.jpg  624_reduced.jpg  625_reduced.jpg  627_reduced.jpg
```

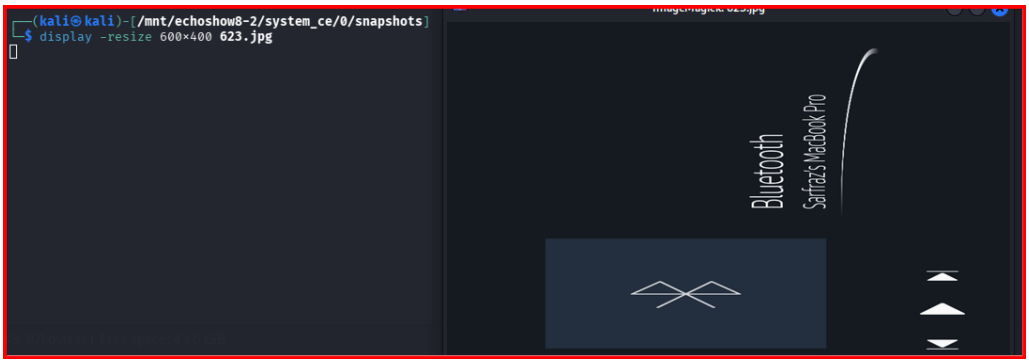
```
┌──(kali㉿kali)-[/mnt/echoshow8-2/system_ce/0/snapshots]
└─$ display -resize 600×400 627.jpg
```



```
┌──(kali㉿kali)-[/mnt/echoshow8-2/system_ce/0/snapshots]
└─$ display -resize 600×400 617.jpg
```




```
┌──(kali㉿kali)-[/mnt/echoshow8-2/system_ce/0/snapshots]
└─$ display -resize 600×400 623.jpg
```




```
┌──(kali㉿kali)-[/mnt/echoshow8-2/system_ce/0]
└─$ cd recent_tasks

┌──(kali㉿kali)-[/mnt/echoshow8-2/system_ce/0/recent_tasks]
└─$ ls
617_task.xml  623_task.xml  624_task.xml  625_task.xml

┌──(kali㉿kali)-[/mnt/echoshow8-2/system_ce/0/recent_tasks]
└─$ cat 617_task.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<task task_id="617" real_activity="com.amazon.cloud9/com.amazon.slate.echo.SlateEchoActivity"
real_activity_suspended="false" affinity="10158:com.amazon.cloud9" root_has_reset="false" auto
_remove_recents="false" asked_compat_mode="false" user_id="0" user_setup_complete="true" effec
tive_uid="10158" last_time_moved="1698022572750" never_relinquish_identity="true" task_descrip
tion_color="ff1f1f1f" task_description_colorBackground="ff1f1f1f" task_affiliation_color="-147
37633" task_affiliation="617" prev_affiliation="-1" next_affiliation="-1" calling_uid="10158"
calling_package="com.amazon.cloud9" resize_mode="1" supports_picture_in_picture="false" min_wi
dth="-1" min_height="-1" persist_task_version="1">
<intent action="android.intent.action.SEND" data="http://awa/search?query=kapil+sharma+show&am
p;provider=youtube&amp;locale=en-US/" component="com.amazon.cloud9/com.amazon.slate.echo.Slate
EchoActivity" flags="14002000" />
</task>

┌──(kali㉿kali)-[/mnt/echoshow8-2/system_ce/0/recent_tasks]
└─$ date -u -d @1698022572
Mon Oct 23 12:56:12 AM UTC 2023
```

BULK EXTRACTOR

A computer forensics tool to extract useful information without parsing the file system or file system structures

- **Some of the features that bulk extractor can extract are: credit card numbers, Internet domains, Email addresses, Ethernet MAC addresses, IP addresses, telephone numbers, etc**
- **The results are stored in the text files**
- **It also creates histograms of features**
- **Very fast**

Create directory and run bulk_extractor command

```
┌──(kali㉿kali)-[/mnt/echoshow8-2]
└─$ cd ~/echoshow8

┌──(kali㉿kali)-[~/echoshow8]
└─$ mkdir bulkext
```

```
┌──(kali㉿kali)-[~/echoshow8]
└─$ bulk_extractor -o bulkext EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin

bulk_extractor version: 2.1.1
Input file: "EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin"
Output directory: "bulkext"
Disk Size: 7818182656
Scanners: aes base64 elf evtx exif facebook find gzip httplogs json kml_carved msxml net ntfsi
ndx ntfslogfile ntfsmft ntfsusn pdf rar sqlite utmp vcard_carved windirs winlnk winpe winprefe
tch zip accts email gps vin
Threads: 4
going multi-threaded ... ( 4 )
bulk_extractor      Mon Oct 12 17:29:42 2026

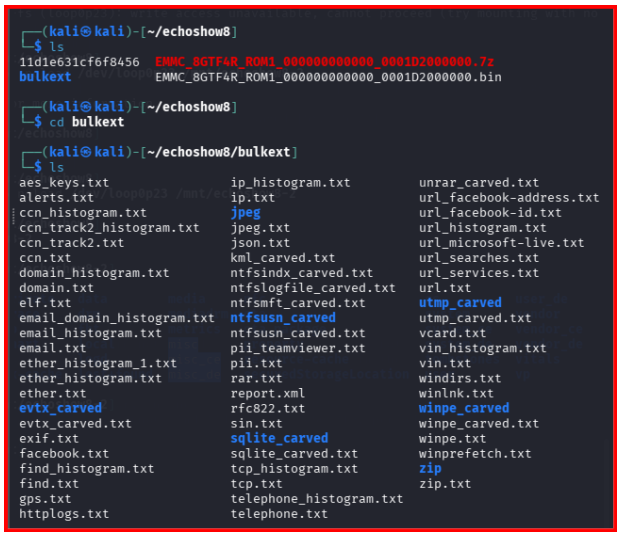
available_memory: 6051287040
bytes_queued: 0
depth0_bytes_queued: 0
depth0_sbufs_queued: 0
elapsed_time:  0:00:00
estimated_date_completion: 2026-10-12 17:29:41
estimated_time_remaining: n/a
fraction_read: 0.000000 %
max_offset: 0
sbufs_created: 0
sbufs_queued: 0
sbufs_remaining: 0
tasks_queued: 0
thread_count: 4
>.....................................................................................|
```

```
bulk_extractor      Mon Oct 12 17:42:29 2026

available_memory: 5722324992
bytes_queued: 63190665
depth0_bytes_queued: 0
depth0_sbufs_queued: 0
elapsed_time:  0:12:47
estimated_date_completion: 2026-10-12 17:42:29
estimated_time_remaining:  0:00:00
fraction_read: 100.000000 %
max_offset: 7801405440
sbufs_created: 38669763
sbufs_queued: 1
sbufs_remaining: 1
tasks_queued: 0
thread-4: 7791411200-GZIP-0: email (63190665 bytes)
thread_count: 4
==========================================================================================>|


Phase 2. Shutting down scanners
Computing final histograms and shutting down ...
Phase 3. Generating stats and printing final usage information
All Threads Finished!
Elapsed time: 768.3 sec.
Total MB processed: 7818
Overall performance: 10.18 MBytes/sec 2.544 (MBytes/sec/thread)
sbufs created:    38669763
sbufs unaccounted: 0
Time producer spent waiting for scanners to process data:         0:10:58 (658.70 seconds)
Time consumer scanners spent waiting for data from producer:      0:01:07 (67.25 seconds)
Average time each consumer spent waiting for data from producer:  0:00:00 (0.00 seconds)
 **** More time spent waiting for workers. You need a faster CPU or more cores for improved perf
ormance.
Total email features found: 18382
```

OUTPUT OF BULK EXTRACTOR

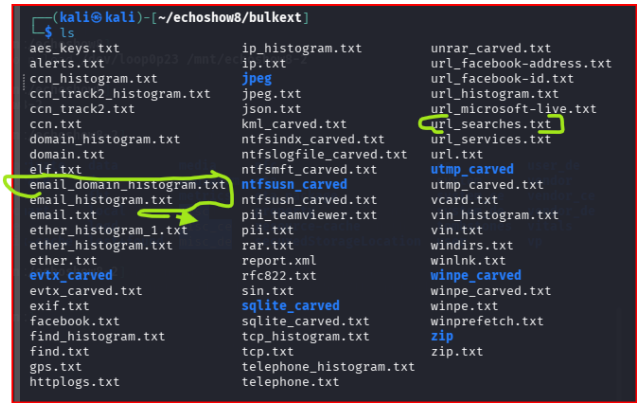


```
┌──(kali㉿kali)-[~/echoshow8]
└─$ ls
11d1e631cf6f8456  EMMC_8GTF4R_ROM1_000000000000_0001D2000000.7z
bulkext           EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin

┌──(kali㉿kali)-[~/echoshow8]
└─$ cd bulkext

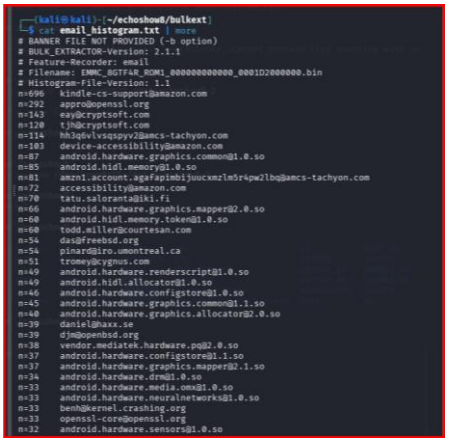
┌──(kali㉿kali)-[~/echoshow8/bulkext]
└─$ ls
aes_keys.txt                  ip_histogram.txt          unrar_carved.txt
alerts.txt                    ip.txt                    url_facebook-address.txt
ccn_histogram.txt             jpeg                      url_facebook-id.txt
ccn_track2_histogram.txt      jpeg.txt                  url_histogram.txt
ccn_track2.txt                json.txt                  url_microsoft-live.txt
ccn.txt                       kml_carved.txt            url_searches.txt
domain_histogram.txt          ntfsindx_carved.txt       url_services.txt
domain.txt                    ntfslogfile_carved.txt    url.txt
elf.txt                       ntfsmft_carved.txt        utmp_carved
email_domain_histogram.txt    ntfsusn_carved            utmp_carved.txt
email_histogram.txt           ntfsusn_carved.txt        vcard.txt
email.txt                     pii_teamviewer.txt        vin_histogram.txt
ether_histogram_1.txt         pii.txt                   vin.txt
ether_histogram.txt           rar.txt                   windirs.txt
ether.txt                     report.xml                winlnk.txt
evtx_carved                   rfc822.txt                winpe_carved
evtx_carved.txt               sin.txt                   winpe_carved.txt
exif.txt                      sqlite_carved             winpe.txt
facebook.txt                  sqlite_carved.txt         winprefetch.txt
find_histogram.txt            tcp_histogram.txt         zip
find.txt                      tcp.txt                   zip.txt
gps.txt                       telephone_histogram.txt
httplogs.txt                  telephone.txt
```

VIEWING EXTRACTED EMAIL ADDRESS.



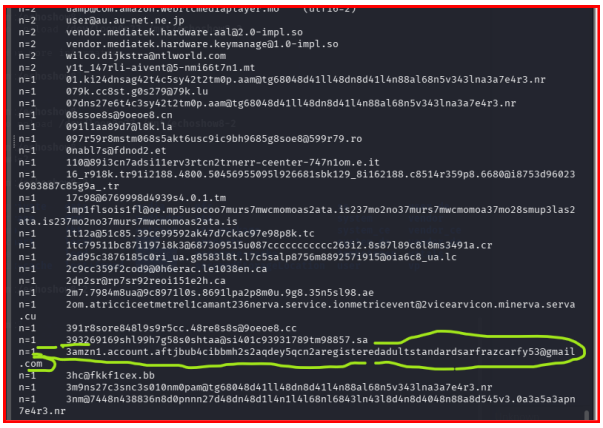
```
┌──(kali㉿kali)-[~/echoshow8/bulkext]
└─$ ls
aes_keys.txt                  ip_histogram.txt          unrar_carved.txt
alerts.txt                    ip.txt                    url_facebook-address.txt
ccn_histogram.txt             jpeg                      url_facebook-id.txt
ccn_track2_histogram.txt      jpeg.txt                  url_histogram.txt
ccn_track2.txt                json.txt                  url_microsoft-live.txt
ccn.txt                       kml_carved.txt            url_searches.txt
domain_histogram.txt          ntfsindx_carved.txt       url_services.txt
domain.txt                    ntfslogfile_carved.txt    url.txt
elf.txt                       ntfsmft_carved.txt        utmp_carved
email_domain_histogram.txt    ntfsusn_carved            utmp_carved.txt
email_histogram.txt           ntfsusn_carved.txt        vcard.txt
email.txt                     pii_teamviewer.txt        vin_histogram.txt
ether_histogram_1.txt         pii.txt                   vin.txt
ether_histogram.txt           rar.txt                   windirs.txt
ether.txt                     report.xml                winlnk.txt
evtx_carved                   rfc822.txt                winpe_carved
evtx_carved.txt               sin.txt                   winpe_carved.txt
exif.txt                      sqlite_carved             winpe.txt
facebook.txt                  sqlite_carved.txt         winprefetch.txt
find_histogram.txt            tcp_histogram.txt         zip
find.txt                      tcp.txt                   zip.txt
gps.txt                       telephone_histogram.txt
httplogs.txt                  telephone.txt
```

VIEWING HISTOGRAM OF EXTRACTED EMAIL ADDRESSES



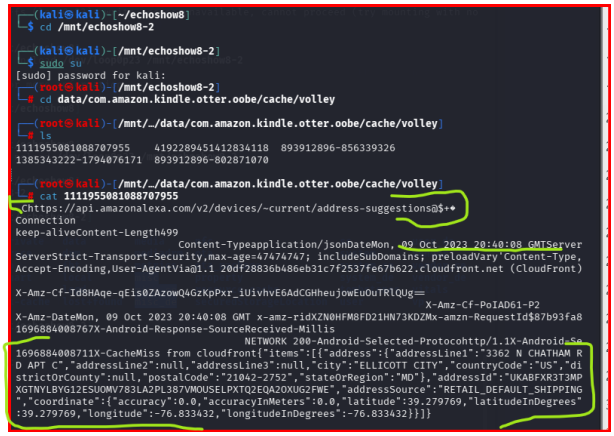
```
┌──(kali㉿kali)-[~/echoshow8/bulkext]
└─$ cat email_histogram.txt | more
# BANNER FILE NOT PROVIDED (-b option)
# BULK_EXTRACTOR-Version: 2.1.1
# Feature-Recorder: email
# Filename: EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
# Histogram-File-Version: 1.1
n=696   kindle-cs-support@amazon.com
n=292   appro@openssl.org
n=143   eay@cryptsoft.com
n=120   tjh@cryptsoft.com
n=114   hh3q6vlvsqspyv2@amcs-tachyon.com
n=103   device-accessibility@amazon.com
n=87    android.hardware.graphics.common@1.0.so
n=85    android.hidl.memory@1.0.so
n=81    amzn1.account.agafapimbijuucxmzlm5r4pw2lbq@amcs-tachyon.com
n=72    accessibility@amazon.com
n=70    tatu.saloranta@iki.fi
n=66    android.hardware.graphics.mapper@2.0.so
n=60    android.hidl.memory.token@1.0.so
n=60    todd.miller@courtesan.com
n=54    das@freebsd.org
n=54    pinard@iro.umontreal.ca
n=51    tromey@cygnus.com
n=49    android.hardware.renderscript@1.0.so
n=49    android.hidl.allocator@1.0.so
n=46    android.hardware.configstore@1.0.so
n=45    android.hardware.graphics.common@1.1.so
n=40    android.hardware.graphics.allocator@2.0.so
n=39    daniel@haxx.se
n=39    djm@openbsd.org
n=38    vendor.mediatek.hardware.pq@2.0.so
n=37    android.hardware.configstore@1.1.so
n=37    android.hardware.graphics.mapper@2.1.so
n=34    android.hardware.drm@1.0.so
n=33    android.hardware.media.omx@1.0.so
n=33    android.hardware.neuralnetworks@1.0.so
n=33    benh@kernel.crashing.org
n=33    openssl-core@openssl.org
n=32    android.hardware.sensors@1.0.so
```

Email Associated with Amazon Account Found!



- The purpose of the app com.amazon.kindle.otter.oobe is to provide a great out-of-box experience (OOBE) for Amazon Fire device
- OOBE is the process of setting up your device for the first time, such as choosing your language, region, Wi-Fi network, and other preferences
- It is used to set up your device settings and preferences during the initial setup process

Physical Address Associated with Amazon Account



The misc/wifi directory contains:

- Files related to the Wi-Fi settings and configuration of the device
- SSID and password of the Wi-Fi networks that device is connected to or saved on the device
- Preferred network

```
┌──(root㉿kali)-[/mnt/echoshow8-2]
└─# cd misc/wifi

┌──(root㉿kali)-[/mnt/echoshow8-2/misc/wifi]
└─# ls
sockets  softap.conf  WifiConfigStore.xml  wpa_supplicant

┌──(root㉿kali)-[/mnt/echoshow8-2/misc/wifi]
└─# cat WifiConfigStore.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<WifiConfigStoreData>
<int name="Version" value="1" />
<NetworkList>
<Network>
<WifiConfiguration>
<string name="ConfigKey">&quot;Fios-pJK9q&quot;WPA_PSK</string>
<string name="SSID">&quot;Fios-pJK9q&quot;</string>
<null name="BSSID" />
<string name="PreSharedKey">&quot;cop576him662ill&quot;</string>
<int name="Priority" value="0" />
<null name="WEPKeys" />
<int name="WEPTxKeyIndex" value="0" />
<boolean name="HiddenSSID" value="false" />
<boolean name="RequirePMF" value="false" />
<byte-array name="AllowedKeyMgmt" num="1">02</byte-array>
<byte-array name="AllowedProtocols" num="1">03</byte-array>
```

WIFI SSID AND PASSWORD

```
<WifiConfiguration>
<string name="ConfigKey">&quot;TellMyWiFiLoveHer&quot;WPA_PSK</string>
<string name="SSID">&quot;TellMyWiFiLoveHer&quot;</string>
<null name="BSSID" />
<string name="PreSharedKey">&quot;cop576him662ill&quot;</string>
<int name="Priority" value="0" />
<null name="WEPKeys" />
<int name="WEPTxKeyIndex" value="0" />
<boolean name="HiddenSSID" value="false" />
<boolean name="RequirePMF" value="false" />
<byte-array name="AllowedKeyMgmt" num="1">02</byte-array>
<byte-array name="AllowedProtocols" num="1">03</byte-array>
<byte-array name="AllowedAuthAlgos" num="1">01</byte-array>
<byte-array name="AllowedGroupCiphers" num="1">0f</byte-array>
<byte-array name="AllowedPairwiseCiphers" num="1">06</byte-array>
<boolean name="Shared" value="true" />
<int name="Status" value="2" />
<null name="FQDN" />
<null name="ProviderFriendlyName" />
<null name="LinkedNetworksList" />
<null name="DefaultGwMacAddress" />
<boolean name="ValidatedInternetAccess" value="true" />
<boolean name="NoInternetAccessExpected" value="false" />
<int name="UserApproved" value="0" />
<boolean name="MeteredHint" value="false" />
<int name="MeteredOverride" value="0" />
<boolean name="UseExternalScores" value="false" />
```

- A system app that is part of the Android Bluetooth stack
- Responsible for managing the Bluetooth functionality on device
- Functions include scanning, pairing, connecting, and transferring data with other Bluetooth devices

BLUETOOTH CONNECTED DEVICES

```
┌──(root㉿kali)-[/mnt/echoshow8-2]
└─# cd user_de/0

┌──(root㉿kali)-[/mnt/echoshow8-2/user_de/0]
└─# cd com.android.bluetooth

┌──(root㉿kali)-[/mnt/echoshow8-2/user_de/0/com.android.bluetooth]
└─# ls
cache  code_cache  lib  shared_prefs

┌──(root㉿kali)-[/mnt/echoshow8-2/user_de/0/com.android.bluetooth]
└─# cd shared_prefs

┌──(root㉿kali)-[/mnt/…/user_de/0/com.android.bluetooth/shared_prefs]
└─# ls
absolute_volume_blacklist.xml       message_access_permission.xml
Bt_Metrics_Device_Connected.xml     phonebook_access_permission.xml
Bt_Metrics_Remote_Device_Uuids.xml  sim_access_permission.xml
Bt_Metrics_Shared_Pref.xml          WirelessMetrics.xml

┌──(root㉿kali)-[/mnt/…/user_de/0/com.android.bluetooth/shared_prefs]
└─# cat Bt_Metrics_Device_Connected.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <boolean name="F4:5C:89:C7:54:F0" value="true" />
</map>
```

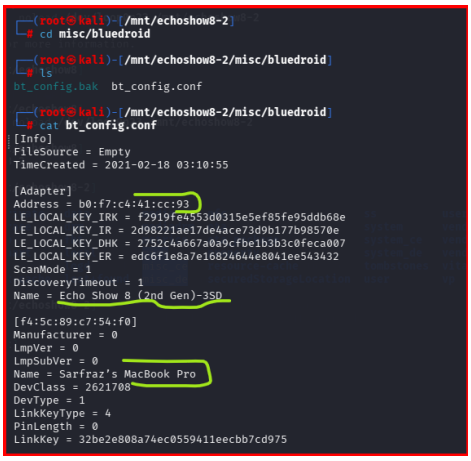
BLUETOOTH PAIRED DEVICES

```
┌──(root㉿kali)-[/mnt/…/user_de/0/com.android.bluetooth/shared_prefs]
└─# cat Bt_Metrics_Remote_Device_Uuids.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <string name="F4:5C:89:C7:54:F0">paired_dev_supp_uuids_a2dpsrc_hfpag_other</string>
    <string name="F4:4E:FD:C6:29:FC">paired_dev_supp_uuids_a2dpsrc_a2dpsink_hfpclient_other</s
tring>
</map>
```

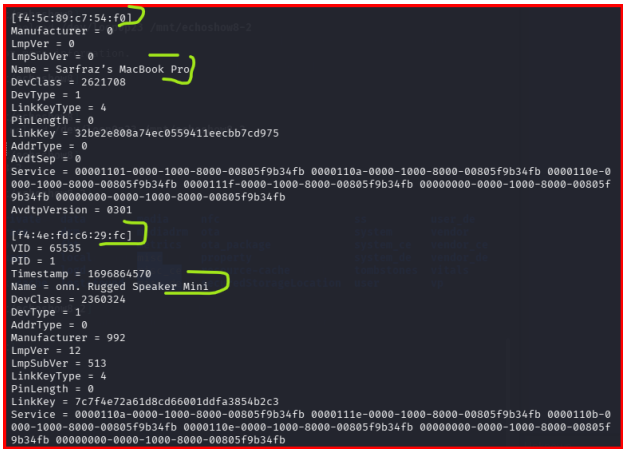
MAC ADDRESS

- A system directory that contains files related to the Bluetooth functionality on the device.
- The bt_config.conf file in this directory stores the Bluetooth configuration data, such as the device name, MAC address, class, etc.
- Name: Device name
- DevClass: It is a 6-digit hexadecimal number that indicates the type and capabilities of your device, such as phone, computer, audio, etc.
- ScanMode: It is a 1-digit decimal number that indicates whether your device is discoverable and connectable by other devices, such as 3 for both, 2 for connectable only, 1 for discoverable only, or 0 for neither.

MAC Address and Name of Echo Show Device



MAC Addresses and Names of Connected Bluetooth Devices

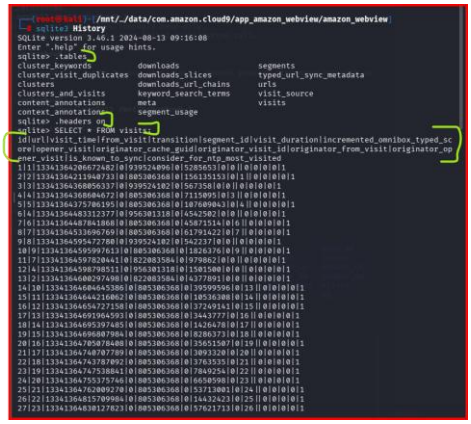


History - Web Browsing and Search History

```
┌──(root㉿kali)-[/mnt/echoshow8-2/data]
└─# cd com.amazon.cloud9/app_amazon_webview/amazon_webview

┌──(root㉿kali)-[/mnt/…/data/com.amazon.cloud9/app_amazon_webview/amazon_webview]
└─# ls
'Account Web Data'                          'Network Action Predictor'
'Account Web Data-journal'                  'Network Action Predictor-journal'
'Affiliation Database'                      'Network Persistent State'
'Affiliation Database-journal'              'Offline Pages'
 AutofillStrikeDatabase                      optimization_guide_hint_cache_store
 blob_storage                                optimization_guide_model_metadata_store
 Bookmarks                                  'Platform Notifications'
 Bookmarks.bak                               Preferences
 BudgetDatabase                              README
 commerce_subscription_db                    RemoteConfiguration-aws_auth
 Cookies                                     RemoteConfiguration-experiments
 Cookies-journal                             RemoteConfiguration-public-devices
 databases                                   RemoteConfiguration-ua-override
 DawnCache                                   RemoteConfiguration-web-scaling-override
 DIPS                                       'Reporting and NEL'
 DIPS-journal                               'Reporting and NEL-journal'
'Download Service'                          'SCT Auditing Pending Reports'
 Favicons                                   'Segmentation Platform'
 Favicons-journal                           'Service Worker'
'Feature Engagement Tracker'                'Session Storage'
 feedv2                                      shared_proto_db
'File System'                                share_history
 GPUCache                                    Shortcuts
 heavy_ad_intervention_opt_out.db            Shortcuts-journal
 heavy_ad_intervention_opt_out.db-journal   'Sync Data'
 History                                    'Top Sites'
 History-journal                            'Top Sites-journal'
 IndexedDB                                   TransportSecurity
'Local Storage'                              VideoDecodeStats
 LOCK                                       'Visited Links'
 LOG                                        'Web Data'
'Login Data'                                'Web Data-journal'
'Login Data-journal'                         WebStorage
 LOG.old
```

BROWSE visits TABLE



Web Searches

TOTAL WEB BROWSING visits



KEYWORDS SEARCHES

```
┌──(root㉿kali)-[/mnt/…/data/com.amazon.cloud9/app_amazon_webview/amazon_webview]
└─# sqlite3 History
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> .headers on
sqlite> .table
cluster_keywords          downloads                 segments
cluster_visit_duplicates  downloads_slices          typed_url_sync_metadata
clusters                  downloads_url_chains      urls
clusters_and_visits       keyword_search_terms      visit_source
content_annotations       meta                      visits
context_annotations       segment_usage
sqlite> SELECT * FROM keyword_search_terms;
keyword_id|url_id|term|normalized_term
3|69|search chicken nuggets on google|search chicken nuggets on google
3|70|close google|close google
sqlite>
```

YOUTUBE SEARCH

```
sqlite> SELECT * FROM urls;
id|url|title|visit_count|typed_count|last_visit_time|hidden
1|https://m.youtube.com/|Home - YouTube|5|0|133422405270715 78|0
2|https://m.youtube.com/shorts/J89×5vmPB2A|Cooking Dinosaur!?! 🦕 - YouTube|2|0|13341364600297498|0
3|https://m.youtube.com/results?search_query=what%27s%20zero%20divided%20by%20one|YouTube|1|0|13341364368056337|0
4|https://m.youtube.com/results?search_query=what%27s+zero+divided+by+one|what's zero divided by one - YouTube|3|0|1
3341364598798511|0
5|https://m.youtube.com/watch?v=NKmGVE85GUU&pp=ygUad2hhdCdzIHplcm8gZGl2aWRlZCBieSBvbmU%3D|Why can't you divide by ze
ro? - TED-Ed - YouTube|1|0|133413643757061 95|0
6|https://m.youtube.com/shorts/dutekN9uuRw|Ask Siri what 0 divided by 0 is ... - YouTube|1|0|133413644878418 68|0
7|https://m.youtube.com/shorts/gClViKU-iDs|YouTube|2|0|133413645978204 41|0
8|https://m.youtube.com/results?search_query=what%27s%20zero%20divided%20by%20zero|YouTube|1|0|13341364595472780|0
9|https://m.youtube.com/results?search_query=what%27s+zero+divided+by+zero|what's zero divided by zero - YouTube|1|0
|13341364595997613|0
10|https://m.youtube.com/shorts/8wxmPWkp8hU|#pov beetle juice tries to convince Lydia to say his name… #acting #beet
lejuice - YouTube|1|0|13341364604645386|0
11|https://m.youtube.com/shorts/uUB2wpH2JgA|This girl starts to sing in the restaurant, EVERYONE WAS SHOCKED 😱 - Yo
uTube|1|0|13341364644216062|0
12|https://m.youtube.com/shorts/umFENSW-zCM|I'm Inside Your House 😱 #shorts #funny #gaming - YouTube|1|0|1334136465
4727158|0
13|https://m.youtube.com/shorts/S3vd3KBm3Vo|I can't believe it made it that far 😱🤯 #shorts - YouTube|1|0|133413646
919645931|0
14|https://m.youtube.com/shorts/aj9AtW_LS4c|Google Translate to Dutch ... - YouTube|1|0|13341364695397485|0
15|https://m.youtube.com/shorts/jTAz9vZtp5Q|the everything restaurant - YouTube|1|0|13341364696807984|0
16|https://m.youtube.com/shorts/hSmdZsaPUlM|WHEN MY KOREAN MOM GETS ROBBED 🤣 - YouTube|1|0|13341364705078408|0
```

SEARCH TABLE URLS

```
47|https://m.youtube.com/watch?v=RgzLnmTaCAU&pp=QAFIBAN3DN3D|Kusu Kusu Song Ft Nora Fatehi | Satyameva Jayate 2 | Jo
hn A, Divya K | Tanishk B, Zahrah Khan, Dev N - YouTube|1|0|13342063812759 00|0
48|https://m.youtube.com/watch?v=IuvVVWOsMBo&pp=QAFIBQN3DN3D|Full Video: Ek Toh Kum Zindagani | Marjaavaan | Nora Fa
tehi | Tanishk B, Neha K, Yash N - YouTube|1|0|13342064052400 275|0
49|https://m.youtube.com/watch?v=IE8OD5F5U-c&pp=QAFIBg%3D%3D|Full Song: Garmi | Street Dancer 3D | Varun D, Nora F,
Badshah, Neha K | Remo D - YouTube|1|0|13342064272865 32|0
50|https://m.youtube.com/results?sp=mAEA&search_query=Kapil+sharma|Kapil sharma - YouTube|2|0|13342240539770 31|0
51|https://m.youtube.com/watch?v=p-ORaVBc5Zc&pp=ygUMS2FwaWwgc2hhcm1h|कपिल शर्मा Kapil धन गंज Yoga Instructor? | The Kapil
 Sharma Show Season 2 | Full Episode - YouTube|2|0|13342240542788213|0
52|https://www.amazon.com/gp/help/customer/display.html?nodeId=201602230|Alexa and Alexa Device FAQs - Amazon Custom
er Service|1|0|13342400688326954|0
53|https://login.microsoftonline.com/common/oauth2/v2.0/authorize?client_id=65ad87c3-6b07-40fa-93e6-e2a734bf2e9f&res
ponse_type=code&state=eyJ2VWxpZGF0aW9uQ29kZSI6ImhUb2dYK2rc3ZPVDl2cTh UbkSkam0=aDVYdWNzRmdRTHBxMEB0V2VGV0U9IiwiZGlyZW
N0ZWR2ZCI6IjhjYzVhYTVmLTEzNWEtOTNlZC1kYjllLTAwMmIxOGYxZTY0ZCIsInBhcnRuZXJJZCI6IndlclNJSWZkSjdhWWd2V1BxZG5mcDgzY0MiLC
JhcHBsaWNhdGlvbkRvbWFpbiI6IkFMRVhBX0hlTyIsImFwcGxpY2F0aW9uRG9tYWlucyI6WyJBTEVYQV9ISE8iXSwiZXhwaXJhdGlvblRpbWVJbk1pbG
xpcyI6MTY5NzkzNzQyOTgzNSwidXNlclN0YXRlIjpbeyJrIjoiYWNjb3VudFVVSUQiLCJ2IjoiOGNjNWFhNWYtMTM1YS05M2VkLWRiOWUtMDAyYjE4Zj
F1MjM6In0seyJrIjoicmVkaXJlY3RUb0FwcCIsInYiOiJmYWxzZSJ9LHsiayI6ImRldmljZUluaXRpYXRlZCIsInYiOiJ0cnVlIn0seyJrIjoibGVnYW
N5Q2FsZW5kYXJBY2NvdW50SWQiLCJ2IjpudWxsfSx7Ims iOiJsb2NhbGUiLCJ2IjoiZW4tdXMifSx7Ims iOiJpbml0aWF0aW5nUmVxdWVzdEklIiwidi
I6IjgzM2JhZmQxLTNlODUtNDI2Yy05ZTZhLTBmODRjZDFjMGYwMy J9LHsiayI6ImZsb3dUeXBlIiwidiI6IkxJTkt JTkcifV0sImRpcmVjdGVkSWRUeX
BlIjoiVVVJRCIsImdyYW50ZWRTY29wZXNJbmNsdWRlZCI6ZmFsc2V9&scope=https%3A%2F%2Fgraph.microsoft.com%2FPeople.Read+https%3
A%2F%2Fgraph.microsoft.com%2Fcontacts.Read+offline_access+https%3A%2F%2Fgraph.microsoft.com%2Fcalendars.Read.Shared+
https%3A%2F%2Fgraph.microsoft.com%2FUser.Read+https%3A%2F%2Fgraph.microsoft.com%2FCalendars.ReadWrite+https%3A%2F%2F
graph.microsoft.com%2FCalendars.ReadWrite.Shared&redirect_uri=https%3A%2F%2Falexa.amazon.com%2Fapi%2F3PAccounts%2Foa
uth%2FestablishAccess&prompt=select_account|Sign in to your account|1|0|13342407432 08665|0
54|https://login.microsoftonline.com/common/oauth2/v2.0/authorize?client_id=65ad87c3-6b07-40fa-93e6-e2a734bf2e9f&res
ponse_type=code&state=eyJ2VWxpZGF0aW9uQ29kZSI6ImNoeG55cWZBYnVJTzZZNDYyVUhWRU1odmxZY1R1eG4rQUhabHMzUE4ZT0E9IiwiZGlyZW
N0ZWR2ZCI6IjgAYzVhYTVmLTljYzktMWY2MC0wODdlLTNjMmQyNmZhYTAzZSIsInBhcnRuZXJJZCI6IndlclNJSWZkSjdhWWd2V1BxZG5mcDgzY0MiLC
JhcHBsaWNhdGlvbkRvbWFpbiI6IkFMRVhBX0hlTyIsImFwcGxpY2F0aW9uRG9tYWlucyI6WyJBTEVYQV9ISE8iXSwiZXhwaXJhdGlvblRpbWVJbk1pbG
xpcyI6MTY5NzkzNzkzNzUwMDIzMCwidXNlclN0YXRlIjpbeyJrIjoiYWNjb3VudFVVSUQiLCJ2IjoiODhjNWFhNWYtOWN jOS0xZjYwLTA4ZUtMCMyZDI2Zm
FhMDNlIn0seyJrIjoicmVkaXJlY3RUb0FwcCIsInYiOiJmYWxzZSJ9LHsiayI6ImRldmljZUluaXRpYXRlZCIsInYiOiJ0cnVlIn0seyJrIjoibGVnYW
N5Q2FsZW5kYXJBY2NvdW50SWQiLCJ2IjpudWxsfSx7Ims iOiJsb2NhbGUiLCJ2IjoiZW4tdXMifSx7Ims iOiJpbml0aWF0aW5nUmVxdWVzdEklIiwidi
```

```
IsImdyYW50ZWRTY29wZXNJbmNsdWRlZCI6ZmFsc2V9|Account Linked|1|0|13342408179474433|0
62|https://account-linking-result.com/en-us/success.html|Account Linked|1|0|13342408179474433|0
63|https://m.youtube.com/results?search_query=kids%20videos|YouTube|1|0|13342410629408781|0
64|https://m.youtube.com/results?search_query=kids+videos|kids videos - YouTube|1|0|13342410630863493|0
65|http://bing.com/|Bing|1|0|13342492001629428|0
66|https://bing.com/|Bing|1|0|13342492001629428|0
67|https://www.bing.com/?toWww=1&redig=FDE7754DDDD74D53AAC684BAF5E19FB3|Bing|2|0|13342492255418343|0
68|https://www.google.com/|Google|2|0|13342492252941224|0
69|https://www.google.com/search?q=search%20chicken%20nuggets%20on%20google|search chicken nuggets on google - Googl
e Search|2|0|13342492250476128|0
70|https://www.google.com/search?q=close%20google|close google - Google Search|1|0|13342492212049982|0
71|https://account.microsoft.com/|Continue|1|0|13342492268854145|0
72|https://login.live.com/login.srf?wa=wsignin1.0&rpsnv=16&checkda=1&ct=1698018668&rver=7.5.2112.0&wp=MBI_SSL&wreply
=https:%2F%2Faccount.microsoft.com%2Fauth%2Fcomplete-silent-signin%3Fru%3Dhttps%253A%252F%252Faccount.microsoft.com%
252F%253Frefd%253Dwww.bing.com&lc=1033&id=292666|Continue|1|0|13342492268854145|0
73|https://account.microsoft.com/?refd=www.bing.com|Microsoft account | Home|2|0|13342492375035788|0
74|https://login.microsoftonline.com/consumers/oauth2/v2.0/authorize?scope=service::account.microsoft.com::MBI_SSL+o
```

```
39|https://www.generationgenius.com/videolessons/human-body-systems-for-kids/?g_adtype=search&g_adgroupid=112753727750&g_campaignid=110954159548g_keyword=human%20body%20systems%20for%20kids&g_network=s&g_adid=652811662799&g_acctid=279-897-8338&g_keywordid=kwd-294924535822&g_campaign=Teachers+-+Sci+-+Topics+-+3-5&utm_source=google&utm_medium=cpc&utm_term=human%20body%20systems%20for%20kids&gclid=EAIaIQobChMIvcvOhPr1gQMVN6JaBR2-zQssEAAYASAAEgKh3_D_BwE|Human Body Systems Video For Kids | 3rd, 4th & 5th Grade|1|0|13341774371057093|0
40|https://www.generationgenius.com/videolessons/human-body-systems-for-kids/|Human Body Systems Video For Kids | 3rd, 4th & 5th Grade|1|0|13341774395750811|0
41|https://m.youtube.com/#searching|Home - YouTube|3|0|13342240529662014|0
42|https://m.youtube.com/results?sp=mAEA&search_query=Hot+indian+music|Hot indian music - YouTube|1|0|13342062897472662|0
43|https://m.youtube.com/watch?v=bi5PhlIQpwU&pp=ygUQSG90IGluZGlhbiBtdXNpYw%3D%3D|Nadiyon Paar - Full Song |Roohi | Janhvi Kapoor| Sachin-Jigar | Shamur | Rashmeet K - YouTube|1|0|13342062940928082|0
44|https://m.youtube.com/watch?v=JFcgOboQZ08&pp=QAFIAQ%3D%3D|DILBAR Lyrical | Satyameva Jayate |John Abraham, Nora F
```

Web Search History – ***Calendar Syn & Microsoft Live Login (email address)***

```
55|https://login.microsoftonline.com/common/oauth2/v2.0/authorize?client_id=65ad87c3-6b07-40fa-93e6-e2a734bf2e9f&response_type=code&state=eyJ2YWxpZGF0aW9uQ29kZSI6ImV3Y3FKRXE2Z3NuWkprODl2VVUzdFlhYWlWUUxsZDFwMGRnamxnRHlqdmM9IiwiZGlyZWN0ZWRJZCI6IjM0YzVhYTY0LTVlY2EtNzUzMi1lYjQxLWRkMDRhODA1OTliMiIsInBhcnRuZXJJZCI6IndlclNJSWZkSjdhWWd2V1BxZG5mcDgzY0MiLCJhcHBsaWNhdGlvbkRvbWFpbiI6IkFMRVhBX0hITyIsImFwcGxpY2F0aW9uRG9tYWlucyI6WyJBTEVYQV9ISE8iXSwiZXhwaXJhdGlvblRpbWVJbk1pbGxpcyI6MTY5NzkzODEyMzgxNywidXNlclN0YXRlIjpbeyJrIjoiYWNjb3VudFVVSUQiLCJ2IjoiMzRjNWFhNjQtNWVjYS03NTMyLWViNDEtZGQwNGE4MDU5OWIyIn0seyJrIjoicmVkaXJlY3RUb0FwcCIsInYiOiJmYWxzZSJ9LHsiayI6ImRldmljZUluaXRpYXRlZCIsInYiOiJ0cnVlIn0seyJrIjoibGVnYWN5Q2FsZW5kYXJBY2NvdW50SWQiLCJ2IjpudWxsfSx7ImsiOiJsb2NhbGUiLCJ2IjoiZW4tdXMifSx7ImsiOiJpbml0aWF0aW5nUmVxdWVzdElkIiwidiI6IkxJTktTJTkciFV0sImRpcmVjdGVkSWRUeXBlIjoiVVVJRCIsImdyYW50ZWRTY29wZXNJbmNsdWRlZCI6ZmFsc2V9&scope=https%3A%2F%2Fgraph.microsoft.com%2FPeople.Read+https%3A%2F%2Fgraph.microsoft.com%2FContacts.Read+offline_access+https%3A%2F%2Fgraph.microsoft.com%2FCalendars.Read.Shared+https%3A%2F%2Fgraph.microsoft.com%2FUser.Read+https%3A%2F%2Fgraph.microsoft.com%2FCalendars.ReadWrite+https%3A%2F%2Fgraph.microsoft.com%2FCalendars.ReadWrite.Shared&redirect_uri=https%3A%2F%2Falexa.amazon.com%2Fapi%2F3PAccounts%2Foauth%2FestablishAccess&prompt=select_account|Sign in to your account|1|0|13342408125501223|0
56|https://login.live.com/oauth20_authorize.srf?client_id=65ad87c3-6b07-40fa-93e6-e2a734bf2e9f&scope=https%3a%2f%2fgraph.microsoft.com%2fPeople.Read+https%3a%2f%2fgraph.microsoft.com%2fContacts.Read+offline_access+https%3a%2f%2fgraph.microsoft.com%2fCalendars.Read.Shared+https%3a%2f%2fgraph.microsoft.com%2fUser.Read+https%3a%2f%2fgraph.microsoft.com%2fCalendars.ReadWrite+https%3a%2f%2fgraph.microsoft.com%2fCalendars.ReadWrite.Shared&redirect_uri=https%3a%2f%2falexa.amazon.com%2fapi%2f3PAccounts%2foauth%2festablishAccess&response_type=code&state=eyJ2YWxpZGF0aW9uQ29kZSI6ImV3Y3FKRXE2Z3NuWkprODl2VVUzdFlhYWlWUUxsZDFwMGRnamxnRHlqdmM9IiwiZGlyZWN0ZWRJZCI6IjM0YzVhYTY0LTVlY2EtNzUzMi1lYjQxLWRkMDRhODA1OTliMiIsInBhcnRuZXJJZCI6IndlclNJSWZkSjdhWWd2V1BxZG5mcDgzY0MiLCJhcHBsaWNhdGlvbkRvbWFpbiI6IkFMRVhBX0hITyIsImFwcGxpY2F0aW9uRG9tYWlucyI6WyJBTEVYQV9ISE8iXSwiZXhwaXJhdGlvblRpbWVJbk1pbGxpcyI6MTY5NzkzODEyMzgxNywidXNlclN0YXRlIjpbeyJrIjoiYWNjb3VudFVVSUQiLCJ2IjoiMzRjNWFhNjQtNWVjYS03NTMyLWViNDEtZGQwNGE4MDU5OWIyIn0seyJrIjoicmVkaXJlY3RUb0FwcCIsInYiOiJmYWxzZSJ9LHsiayI6ImRldmljZUluaXRpYXRlZCIsInYiOiJ0cnVlIn0seyJrIjoibGVnYWN5Q2FsZW5kYXJBY2NvdW50SWQiLCJ2IjpudWxsfSx7ImsiOiJsb2NhbGUiLCJ2IjoiZW4tdXMifSx7ImsiOiJpbml0aWF0aW5nUmVxdWVzdElkIiwidiI6IkxJTktTJTkciFV0sImRpcmVjdGVkSWRUeXBlIjoiVVVJRCIsImdyYW50ZWRTY29wZXNJbmNsdWRlZCI6ZmFsc2V9&uaid=0b1ceb5b95c84f158c415a49d7f278ab&msproxy=1&issuer=mso&tenant=common&ui_locales=en-US&epct=PAQABAAEAAAAtyolDObpQQ5VtlI4uGjEPm5VTyAIXMgTbGgYtRtr2rlkkU61OldaZswJtn27_tmDIVdYUPWfN5NOxGKEaEEumjHVutz_4VBM2eCWkSbnicwswuBRZKiQu7jC_kLpeNqFlhmkKQINcsuQBraMR1ZCAqtZJTh557Urr5eayExnlF-NWyQ9cPOLI4Tw8VxuvCZE1tyvmBsp7eu7qfeG3Ldg_FkyRVxnehcUtEv2TPnF8QyAA&jshs=0&username=sarfy53%40hotmail.com&login_hint=sarfy53%40hotmail.com|Sign in to your Microsoft account|5|0|13342408145560 2|0
57|https://login.live.com/ppsecure/post.srf?username=sarfy53%40hotmail.com&client_id=65ad87c3-6b07-40fa-93e6-e2a734bf2e9f&contextid=9FF24BA9C6C65811&opid=1D443372257DCC7C&bk=1697934545&uaid=0b1ceb5b95c84f158c415a49d7f278ab&pid=15216|Microsoft account|5|0|13342408159065431|0
58|https://login.live.com/ppsecure/post.srf?username=sarfy53%40hotmail.com&client_id=65ad87c3-6b07-40fa-93e6-e2a734b
```

PHONE COMMUNICATION

- Phone communication logs are saved in XML files.
- However, these XML files are deleted by the operating system (OS).
- As a result, these files cannot be found in the file system.
- Fortunately, they can be carved from the unallocated space.
- A popular file-carving tool in Linux for this purpose is the 'scalpel.'
- To use scalpel effectively, we need to specify the type of files we want to carve by editing its configuration file.

```
┌──(kali㉿kali)-[~/echoshow8]
└─$ mmls  EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Safety Table
001:  ------    0000000000   0000000063   0000000064   Unallocated
002:  Meta      0000000001   0000000001   0000000001   GPT Header
003:  Meta      0000000002   0000000033   0000000032   Partition Table
004:  000       0000000064   0000002111   0000002048   kb
005:  001       0000002112   0000004159   0000002048   dkb
006:  002       0000004160   0000014399   0000010240   tee1
007:  003       0000014400   0000032767   0000018368   tee2
008:  004       0000032768   0000043007   0000010240   persist
009:  005       0000043008   0000044031   0000001024   misc
010:  006       0000044032   0000125951   0000081920   recovery
011:  007       0000125952   0000191487   0000065536   swdl
012:  008       0000191488   0000272383   0000080896   metadata
013:  009       0000272384   0000274431   0000002048   boot_para
014:  010       0000274432   0000294911   0000020480   nvcfg
015:  011       0000294912   0000880639   0000585728   vendor
016:  012       0000880640   0000882687   0000002048   spmfw
017:  013       0000882688   0000884735   0000002048   sspm_1
018:  014       0000884736   0000915455   0000030720   cam_vpu1
019:  015       0000915456   0000980991   0000065536   cam_vpu2
020:  016       0000980992   0001011711   0000030720   cam_vpu3
021:  017       0001011712   0001013759   0000002048   lk
022:  018       0001013760   0001015807   0000002048   lk2
023:  019       0001015808   0001048575   0000032768   boot
024:  020       0001048576   0007405567   0006356992   system
025:  021       0007405568   0007929855   0000524288   cache
026:  022       0007929856   0015269854   0007339999   userdata
027:  ------    0015269855   0015269887   0000000033   Unallocated
```

dd if=EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin \ of=userdata_partition.bin \ bs=512 \ skip=7929856 \count=7339999

```
┌──(kali㉿kali)-[~/echoshow8]
└─$ dd if=EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin \
   of=userdata_partition.bin \
   bs=512 \
   skip=7929856 \
   count=7339999

7339999+0 records in
7339999+0 records out
3758079488 bytes (3.8 GB, 3.5 GiB) copied, 90.9869 s, 41.3 MB/s
```

Isolating Userdata Partition from Disk Image

```
┌──(kali㉿kali)-[~/echoshow8]
└─$ ll
total 13644816
-rw-rw-r-- 1 kali kali       5877 Jul  6 23:16 11d1e631cf6f8456
drwxrwxr-x 9 kali kali       4096 Oct 12 17:42 bulkext
-rw-rw-r-- 1 kali kali 2396000768 Oct 11 09:18 EMMC_8GTF4R_ROM1_000000000000_0001D2000000.7z
-rw-rw-r-- 1 kali kali 7818182656 Dec 29  2023 EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
-rw-rw-r-- 1 kali kali 3758079488 Oct 14 10:11 userdata_partition.bin

┌──(kali㉿kali)-[~/echoshow8]
└─$ ls -lh userdata_partition.bin
-rw-rw-r-- 1 kali kali 3.5G Oct 14 10:11 userdata_partition.bin
```

Carving XML Files – Edit & Save Conf. File

```
┌──(kali㉿kali)-[~/echoshow8]
└─$ mkdir recovered-xml

┌──(kali㉿kali)-[~/echoshow8]
└─$ sudo nano /etc/scalpel/scalpel.conf
[sudo] password for kali:
```

```
#
#---------------------------------------------------------
# ScanSoft PaperPort "Max" files
#---------------------------------------------------------
#      max    y      1000000     \x56\x69\x47\x46\x6b\x1a\x00\x00\x00\x00   \x00\x00\x05\x80\x00\x00
#---------------------------------------------------------
# PINs Password Manager program
#---------------------------------------------------------
#      pins   y      8000      \x50\x49\x4e\x53\x20\x34\x2e\x32\x30\x0d
       xml    y     20000      <?xml
```

Carving XML Files – Carving Files

```
┌──(kali㉿kali)-[~/echoshow8]
└─$ scalpel -o recovered-xml userdata_partition.bin
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/home/kali/echoshow8/userdata_partition.bin"

Image file pass 1/2.
userdata_partition.bin: 100.0% |************************************************************|    3.5 GB    00:00 ETA
Allocating work queues ...
Work queues allocation complete. Building carve lists ...
Carve lists built.  Workload:
xml with header "\x3c\x3f\x78\x6d\x6c" and footer "" ⟶ 12205 files
Carving files from image.
Image file pass 2/2.
userdata_partition.bin: 100.0% |************************************************************|    3.5 GB    00:00 ETA
Processing of image file complete. Cleaning up ...
Done.
Scalpel is done, files carved = 12205, elapsed = 10 seconds.
```

Go the output directory and view the recovered file

```
┌──(kali㉿kali)-[~/echoshow8]
└─$ cd recovered-xml

┌──(kali㉿kali)-[~/echoshow8/recovered-xml]
└─$ ls
audit.txt  xml-0-1   xml-0-11  xml-0-2  xml-0-4  xml-0-6  xml-0-8
xml-0-0    xml-0-10  xml-0-12  xml-0-3  xml-0-5  xml-0-7  xml-0-9

┌──(kali㉿kali)-[~/echoshow8/recovered-xml]
└─$ cat audit.txt

Scalpel version 1.60 audit file
Started at Wed Oct 14 10:37:38 2026
Command line:
scalpel -o recovered-xml userdata_partition.bin

Output directory: /home/kali/echoshow8/recovered-xml
Configuration file: /etc/scalpel/scalpel.conf

Opening target "/home/kali/echoshow8/userdata_partition.bin"

The following files were carved:
File              Start              Chop           Length          Extracted From
00000000.xml     12582912              YES          20000            userdata_partition.bin
00000001.xml    449131116              YES          20000            userdata_partition.bin
00000007.xml    529514256              YES          20000            userdata_partition.bin
00000006.xml    529487998              YES          20000            userdata_partition.bin
00000005.xml    529484969              YES          20000            userdata_partition.bin
00000004.xml    529484630              YES          20000            userdata_partition.bin
00000003.xml    529461363              YES          20000            userdata_partition.bin
00000002.xml    529447968              YES          20000            userdata_partition.bin
00000015.xml    551074501              YES          20000            userdata_partition.bin
00000014.xml    551074473              YES          20000            userdata_partition.bin
```

Searching for Callee Name ‘hibna’ in carved XML Files

```
┌──(kali㉿kali)-[~/echoshow8/recovered-xml]
└─$ grep -Eril 'hibna.*2023-10-21|2023-10-21.*hibna'
xml-0-8/00008575.xml
xml-0-8/00008573.xml
xml-0-8/00008574.xml
xml-0-8/00008570.xml
xml-0-8/00008571.xml
xml-0-7/00007278.xml
xml-0-7/00007274.xml
xml-0-7/00007288.xml
xml-0-7/00007276.xml
xml-0-7/00007289.xml
xml-0-7/00007277.xml
xml-0-7/00007285.xml
xml-0-7/00007279.xml
xml-0-7/00007284.xml
xml-0-7/00007275.xml
```

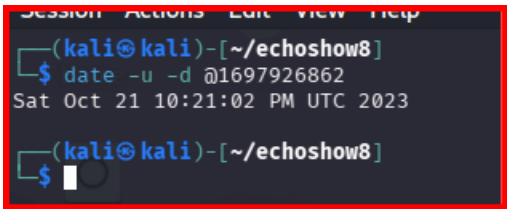
View Files Containing the Callee Name ‘hibna’

```
┌──(kali㉿kali)-[~/echoshow8/recovered-xml]
└─$ grep -Ei 'hibna.*2023-10-21|2023-10-21.*hibna' xml-0-8/00008575.xml
    <string name="SipClient:SipClientState">{&quot;@class&quot;:&quot;amazon.speech.simclient.context.persistence.DeviceContextSerializable&quot;,&quot;caller&quot;:&quot;com.amazon.comms.multimodaltachyonarm&quot;,&quot;endpointId&quot;:null,&quot;instance&quot;:null,&quot;lifecycle&quot;:2,&quot;name&quot;:&quot;SipClientState&quot;,&quot;namespace&quot;:&quot;SipClient&quot;,&quot;payload&quot;:&quot;{\&quot;CLIENT_IDENTIFIER\&quot;:\&quot;Domain:Application:Communications:Calling\&quot;,\&quot;deviceSentAbsoluteTimestamp\&quot;:\&quot;1697926862078\&quot;,\&quot;deviceSentRelativeTimestamp\&quot;:\&quot;764771342\&quot;,\&quot;sipCallState\&quot;:{\&quot;callActive\&quot;:null,\&quot;callConnectDurationInMillis\&quot;:0,\&quot;callId\&quot;:\&quot;5c32191e-c27f-472e-a81f-50e7dce5788f\&quot;,\&quot;callProvider\&quot;:\&quot;Bacon\&quot;,\&quot;callState\&quot;:\&quot;CALLING\&quot;,\&quot;callTotalDurationInMillis\&quot;:0,\&quot;callee\&quot;:{\&quot;id\&quot;:\&quot;\&quot;,\&quot;name\&quot;:\&quot;&lt;scrub&gt;Hibna Sarfraz&lt;/scrub&gt;\&quot;,\&quot;sipInstance\&quot;:null,\&quot;uri\&quot;:\&quot;sips:AAAAAAAAAACO60J6r+eHauADAb4X4eQkKAAAAAAAAACh1rUdIEDXbv+OrsThM2oNOuQG2RVyWo+LYJyzvracOh6tzrE87lmH@amcs-tachyon.com;user=phone\&quot;},\&quot;caller\&quot;:{\&quot;id\&quot;:\&quot;amzn1.comms.id.hg.amzn1~HH3Q6VLVSQSPYV2\&quot;,\&quot;name\&quot;:\&quot;&lt;scrub&gt;&lt;/scrub&gt;\&quot;,\&quot;sipInstance\&quot;:null,\&quot;uri\&quot;:\&quot;sips:id.hg.amzn1~HH3Q6VLVSQSPYV2@amcs-tachyon.com\&quot;},\&quot;isDropIn\&quot;:false,\&quot;side\&quot;:\&quot;caller\&quot;,\&quot;sipCallId\&quot;:null,\&quot;sipRegistered\&quot;:true}}&quot;,&quot;publishState&quot;:&quot;UNPUBLISHED&quot;,&quot;timeOfSample&quot;:&quot;2023-10-21T22:21:02.100Z&quot;,&quot;timestamp&quot;:1697926862100,&quot;uncertaintyInMillis&quot;:0,&quot;version&quot;:100}</string>
```

Phone log found



Convert the epoch time UTC and EST



MULTIMEDIA

PHONE (IMAGES)

Three types

- Captured by the device camera
- Downloaded from the cloud (Amazon Cloud)
- Downloaded during activities like playing music, videos, and
- None of these is saved on device file system

- Conduct a search for file type jpg in user data partition and check

```
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~/echoshow8]
└─$ mount | grep echoshow8-2

┌──(kali㉿kali)-[~/echoshow8]
└─$ losetup -a

┌──(kali㉿kali)-[~/echoshow8]
└─$ sudo losetup --find --show userdata_partition.bin
[sudo] password for kali:
/dev/loop0

┌──(kali㉿kali)-[~/echoshow8]
└─$ sudo blkid /dev/loop1

┌──(kali㉿kali)-[~/echoshow8]
└─$ file -s /dev/loop1
/dev/loop1: no read permission
```

```
┌──(kali㉿kali)-[~/echoshow8]
└─$ losetup -a
/dev/loop0: []: (/home/kali/echoshow8/userdata_partition.bin)

┌──(kali㉿kali)-[~/echoshow8]
└─$ sudo blkid /dev/loop0
/dev/loop0: LABEL="data" UUID="675ea112-677b-40da-8ab3-de348e4712ea" BLOCK_SIZE="4096" TYPE="ext4"

┌──(kali㉿kali)-[~/echoshow8]
└─$ sudo mount -o ro,noload /dev/loop0 /mnt/echoshow8-2

┌──(kali㉿kali)-[~/echoshow8]
└─$ ls -lh /mnt/echoshow8-2
total 196K
drwx------.   2 root root 4.0K Dec 31  2009 adb
drwxrwxr-x.   2 kali kali 4.0K Oct 21  2023 anr
drwxrwx--x.  17 kali kali 4.0K Oct 17  2023 app
drwx------.   2 root root 4.0K Dec 31  2009 app-asec
drwxrwx--x.   2 kali kali 4.0K Dec 31  2009 app-ephemeral
drwxrwx--x.   2 kali kali 4.0K Dec 31  2009 app-lib
drwxrwx--x.   2 kali kali 4.0K Dec 31  2009 app-private
drwxrwx--x.   2 root kali 4.0K Dec 31  2009 autotrace
drwx------.   4 kali kali 4.0K Oct 22  2023 backup
drwxr-xr-x.   2 2000 2000 4.0K Dec 31  2009 bootchart
drwxrwx---.   5 kali 2001 4.0K Dec 31  2009 cache
drwxrwx--x.   3 root root 4.0K Dec 31  2009 dalvik-cache
drwxrwx--x. 203 kali kali  16K Oct  3  2023 data
drwxrwx---.   3 1019 1019 4.0K Dec 31  2009 drm
drwxrwx---.   2 1005 1005 4.0K Oct  3  2023 fht
```

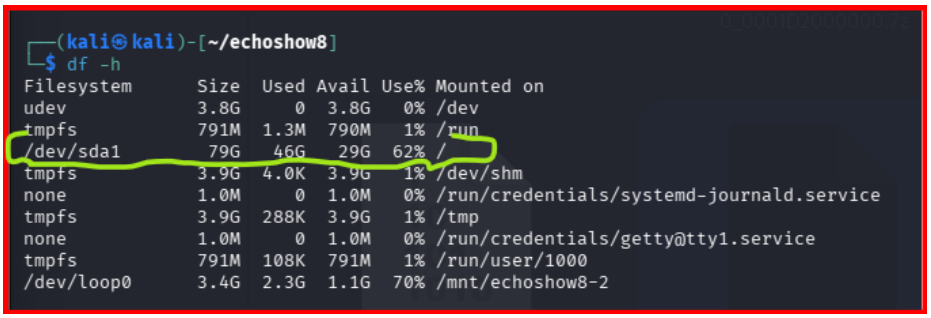
#

Found 12 .jpg (*sudo find /mnt/echoshow8-2 -type f -iname "*.jpg" -print)*

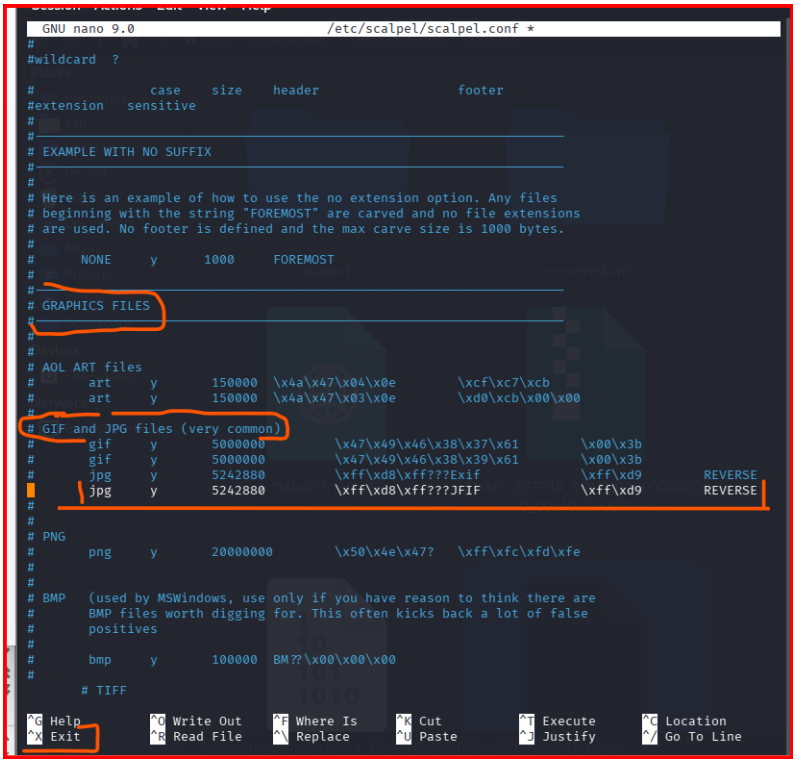
```
┌──(kali㉿kali)-[~/echoshow8]
└─$ sudo find /mnt/echoshow8-2 -type f -iname "*.jpg" -print
/mnt/echoshow8-2/media/0/deviceConfigBundles/img_06_skills.jpg
/mnt/echoshow8-2/media/0/extractedTar/2/img_06_skills.jpg
/mnt/echoshow8-2/system_ce/0/snapshots/623_reduced.jpg
/mnt/echoshow8-2/system_ce/0/snapshots/627_reduced.jpg
/mnt/echoshow8-2/system_ce/0/snapshots/623.jpg
/mnt/echoshow8-2/system_ce/0/snapshots/624.jpg
/mnt/echoshow8-2/system_ce/0/snapshots/617.jpg
/mnt/echoshow8-2/system_ce/0/snapshots/625.jpg
/mnt/echoshow8-2/system_ce/0/snapshots/625_reduced.jpg
/mnt/echoshow8-2/system_ce/0/snapshots/627.jpg
/mnt/echoshow8-2/system_ce/0/snapshots/617_reduced.jpg
/mnt/echoshow8-2/system_ce/0/snapshots/624_reduced.jpg
```

- Only 12 jpg files found in userdata partition
- No jpg files in system partition
- No photos captured by the device camera, downloaded from the cloud or downloaded from internet found in the file systme
- But, the photos can be recovered through data carving techniques

We proceed to DATA CARVING



Carving Images(jpg files) withv ‘scalpel’



RUN SCAPEL COMMAND

Scalpel EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin

```
┌──(kali㉿kali)-[~/echoshow8]
└─$ sudo nano /etc/scalpel/scalpel.conf
[sudo] password for kali:

┌──(kali㉿kali)-[~/echoshow8]
└─$ sudo nano /etc/scalpel/scalpel.conf

┌──(kali㉿kali)-[~/echoshow8]
└─$ scalpel EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/home/kali/echoshow8/EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin"

Image file pass 1/2.
EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin: 100.0% |******************|    7.3 GB    00:00 ETA
Allocating work queues ...
Work queues allocation complete. Building carve lists ...
Carve lists built.  Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" ⟶ 6393 files
xml with header "\x3c\x3f\x78\x6d\x6c" and footer "" ⟶ 12757 files
Carving files from image.
Image file pass 2/2.
EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin: 100.0% |******************|    7.3 GB    00:00 ETA
Processing of image file complete. Cleaning up ...
Done.
Scalpel is done, files carved = 19150, elapsed = 378 seconds.
```

```
┌──(kali㉿kali)-[~/echoshow8]
└─$ ls
11d1e631cf6f8456                              recovered-xml
bulkext                                       scalpel-output
EMMC_8GTF4R_ROM1_000000000000_0001D2000000.7z userdata_partition.bin
EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin

┌──(kali㉿kali)-[~/echoshow8]
└─$ cd scalpel-output

┌──(kali㉿kali)-[~/echoshow8/scalpel-output]
└─$ ls
audit.txt  jpg-0-2  jpg-0-5  xml-1-1   xml-1-12  xml-1-4  xml-1-7
jpg-0-0    jpg-0-3  jpg-0-6  xml-1-10  xml-1-2   xml-1-5  xml-1-8
jpg-0-1    jpg-0-4  xml-1-0  xml-1-11  xml-1-3   xml-1-6  xml-1-9

┌──(kali㉿kali)-[~/echoshow8/scalpel-output]
└─$ cd jpg-0-3

┌──(kali㉿kali)-[~/echoshow8/scalpel-output/jpg-0-3]
└─$ ls | head
00003000.jpg
00003001.jpg
00003002.jpg
00003003.jpg
00003004.jpg
00003005.jpg
00003006.jpg
00003007.jpg
00003008.jpg
00003009.jpg

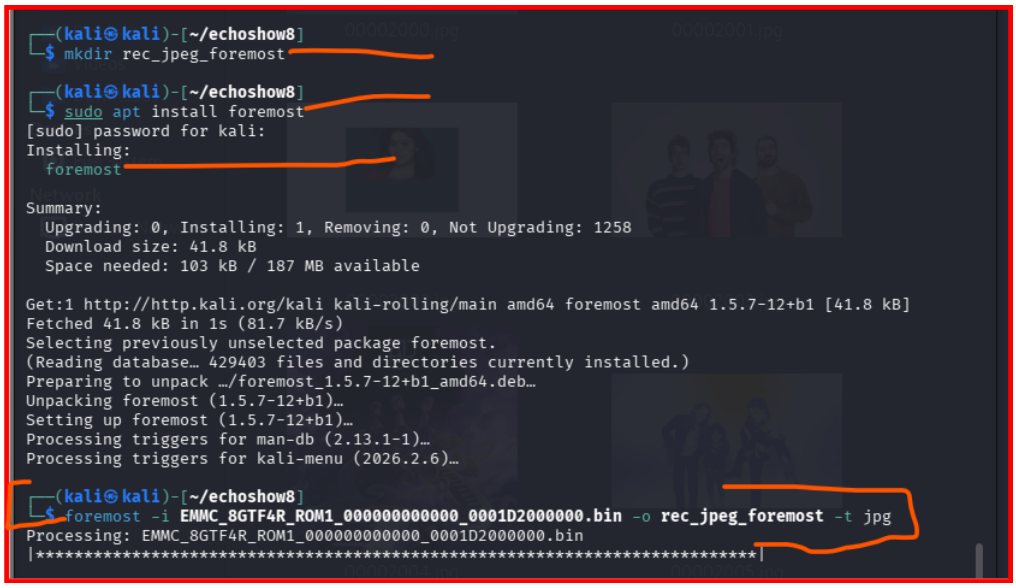
┌──(kali㉿kali)-[~/echoshow8/scalpel-output/jpg-0-3]
└─$ display -resize 00003000.jpg
display: invalid argument for option '-resize': 00003000.jpg @ er
/1545.

┌──(kali㉿kali)-[~/echoshow8/scalpel-output/jpg-0-3]
└─$ display -resize 600×400  00003000.jpg
```

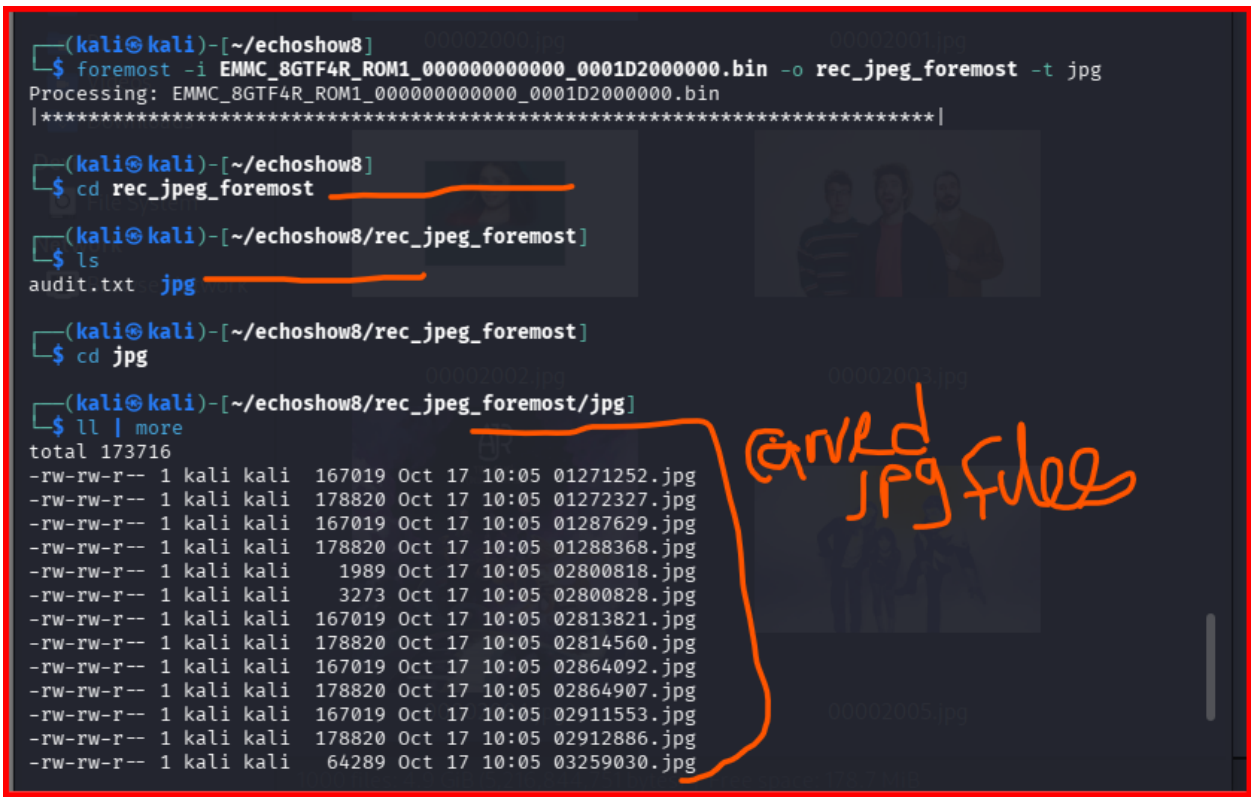


- Scalpel produces unnecessary large output
- More than 27 GB for a 7.3 GB image file
- Same images are carved multiple time
- *Foremost* is a better, easier, and faster option

TASK DONE: FOREMOST INSTALLED TO CONTINUE (sudo apt install foremost)



Carving images with *foremost (view output of foremost)*



View audit report of foremost

```
┌──(kali㉿kali)-[~/echoshow8/rec_jpeg_foremost/jpg]
└─$ cd ..

┌──(kali㉿kali)-[~/echoshow8/rec_jpeg_foremost]
└─$ ls
audit.txt  jpg

┌──(kali㉿kali)-[~/echoshow8/rec_jpeg_foremost]
└─$ cat audit.txt
Foremost version 1.5.7 by Jesse Kornblum, Kris Kendall, and Nick Mikus
Audit File

Foremost started at Sat Oct 17 10:05:53 2026
Invocation: foremost -i EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin -o rec_jpeg_foremost -t jpg

Output directory: /home/kali/echoshow8/rec_jpeg_foremost
Configuration file: /etc/foremost.conf
------------------------------------------------------------------
File: EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
Start: Sat Oct 17 10:05:53 2026
Length: 7 GB (7818182656 bytes)

Num      Name (bs=512)         Size      File Offset     Comment

0:      01271252.jpg         163 KB       650881084
1:      01272327.jpg         174 KB       651431844
2:      01287629.jpg         163 KB       659266088
3:      01288368.jpg         174 KB       659644812
4:      02800818.jpg           1 KB      1434018976
5:      02800828.jpg           3 KB      1434023960
6:      02813821.jpg         163 KB      1440676760
7:      02814560.jpg         174 KB      1441054896
8:      02864092.jpg         163 KB      1466415248
9:      02864907.jpg         174 KB      1466832620
10:     02911553.jpg         163 KB      1490715632
```

```
6384:   15147312.jpg          42 KB      7755423744
6385:   15147400.jpg          54 KB      7755468800
6386:   15147512.jpg          59 KB      7755526144
6387:   15147632.jpg          36 KB      7755587584
6388:   15147712.jpg          74 KB      7755628544
6389:   15147864.jpg          52 KB      7755706368
Finish: Sat Oct 17 10:06:18 2026

6390 FILES EXTRACTED

jpg:= 6390
------------------------------------------------------------------

Foremost finished at Sat Oct 17 10:06:18 2026
```

Carving Images – *Image taken by Camera*

```
6389:   15147864.jpg          52 KB      7755706368
Finish: Sat Oct 17 10:06:18 2026

6390 FILES EXTRACTED

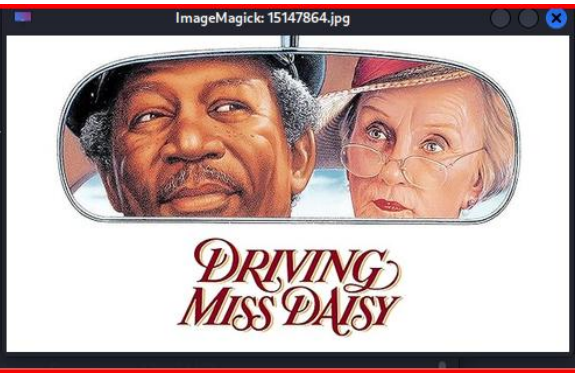
jpg:= 6390
------------------------------------------------------------------

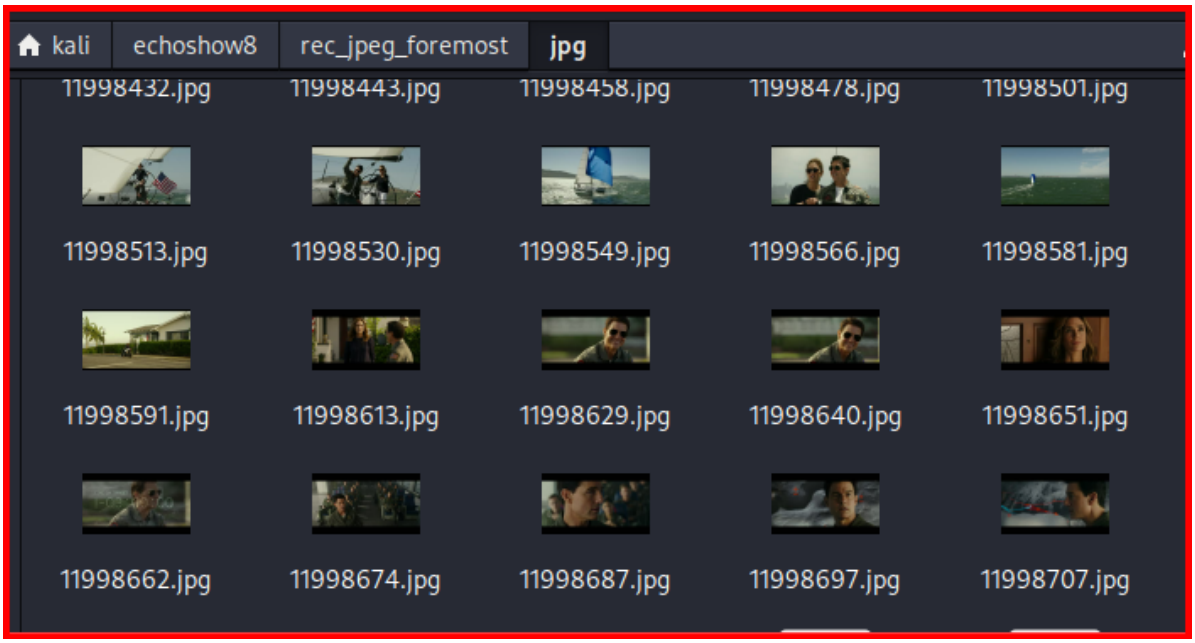
Foremost finished at Sat Oct 17 10:06:18 2026

┌──(kali㉿kali)-[~/echoshow8/rec_jpeg_foremost]
└─$ cd jpg

┌──(kali㉿kali)-[~/echoshow8/rec_jpeg_foremost/jpg]
└─$ display -resize 600×400   10796728.jpg

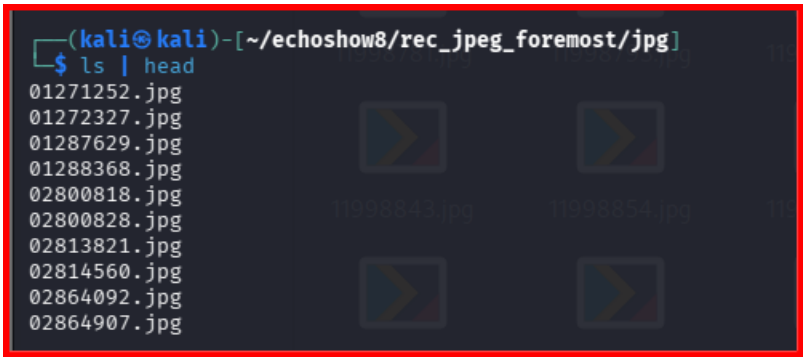
┌──(kali㉿kali)-[~/echoshow8/rec_jpeg_foremost/jpg]
└─$ display -resize 600×400   15147864.jpg
```





Metadata of Recovered Photos (jpg)

Metadata of Carved Photos (jpg)



Use exiftool to view metadata

```
┌──(kali㉿kali)-[~/echoshow8/rec_jpeg_foremost/jpg]
└─$ exiftool 01271252.jpg
ExifTool Version Number         : 13.55
File Name                       : 01271252.jpg
Directory                       : .
File Size                       : 167 kB
File Modification Date/Time     : 2026:10:17 10:05:55-04:00
File Access Date/Time           : 2026:10:17 10:37:40-04:00
File Inode Change Date/Time     : 2026:10:17 10:05:55-04:00
File Permissions                : -rw-rw-r--
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
JFIF Version                    : 1.01
Resolution Unit                 : None
X Resolution                    : 72
Y Resolution                    : 72
Image Width                     : 480
Image Height                    : 480
Encoding Process                : Progressive DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:4:4 (1 1)
Image Size                      : 480×480
Megapixels                      : 0.230
```

- **Carving tools, like foremost, extract raw file content during recovery.**
- **Metadata, such as creation date, may be lost.**

Videos Played on Device

- File *dbplaybackhistory.db* contains information related to videos played
- File is in package com.amazon.avod
- Exact location is data/com.amazon.avod/files/databases/dbplaybackhistory

```
┌──(kali㉿kali)-[/mnt/echoshow8-2]
└─$ sudo su
[sudo] password for kali:
┌──(root㉿kali)-[/mnt/echoshow8-2]
└─# cd data/com.amazon.avod/files/databases

┌──(root㉿kali)-[/mnt/…/data/com.amazon.avod/files/databases]
└─# ls
bookmark      bookmark-wal      event      event-wal     mediaEvents-shm
bookmark-shm  dbplaybackhistory event-shm  mediaEvents   mediaEvents-wal
```

Uploading Data of Photos and Videos Captured by Device – *06 photos & 02 videos uploaded*

```
┌──(kali㉿kali)-[/mnt/echoshow8-2/data/com.amazon.zordon/databases]
└─$ sudo su
[sudo] password for kali:
┌──(root㉿kali)-[/mnt/echoshow8-2/data/com.amazon.zordon/databases]
└─# ls
amzn1.account.AFTJBUB4CIBBMH2S2AQDEY5QCN2A.mixtape.db
amzn1.account.AFTJBUB4CIBBMH2S2AQDEY5QCN2A.mixtape.db-shm
amzn1.account.AFTJBUB4CIBBMH2S2AQDEY5QCN2A.mixtape.db-wal
arn:aws:remote-config:us-west-2:129701941857:appConfig:acg0qhzz_ConfigurationCache.db
arn:aws:remote-config:us-west-2:294587416475:appConfig:azg8ownl_ConfigurationCache.db
arn:aws:remote-config:us-west-2:357484902646:appConfig:a51zegux_ConfigurationCache.db
arn:aws:remote-config:us-west-2:357484902646:appConfig:am7rz1w4_ConfigurationCache.db
mixtape_accounts.db
mixtape_accounts.db-shm
mixtape_accounts.db-wal

┌──(root㉿kali)-[/mnt/echoshow8-2/data/com.amazon.zordon/databases]
└─# sqlite3 amzn1.account.AFTJBUB4CIBBMH2S2AQDEY5QCN2A.mixtape.db
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> .tables
account_metadata_states        family_member_roles
android_metadata               family_members
checkpoints                    metadata_cleanup_queue
cloud_media_parents            node_children_kind_view
cloud_node_children            node_groups
cloud_node_cover_photos        node_kinds
cloud_node_labels              node_kinds_view
cloud_node_parents             node_search_tags
cloud_node_properties          node_statuses
cloud_node_query_excluded_ids  node_transforms
cloud_node_query_include       notification_topic_subscriptions
cloud_node_transforms          pending_migrations
cloud_node_transforms_view     things
cloud_nodes                    upload_log
document_authors               upload_queue_blockers
events                         upload_request_blockers
family_ids                     upload_requests
sqlite>
```

06 photos and 02 videos uploaded

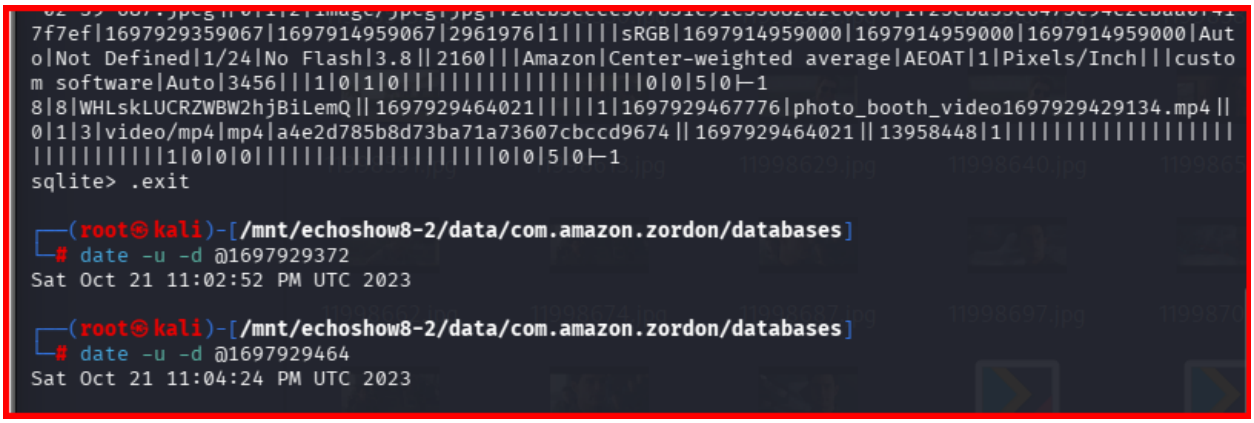
```
sqlite> .header on
sqlite> SELECT * FROM upload requests;
Parse error: no such table: upload
sqlite> SELECT * FROM upload_requests;
_id|queue|parent_id|file_uri|file_size|name|priority|state|suppress_deduplication|rename_on_name_c
onflict|retry_exceptions|last_error|attempts|md5|local_id|node_id|bytes_sent|bytes_received|catego
ry|cds_path|app_data|content_date|add_to_family_archive|created_date|visual_digest
1|photo_booth|5GxpwT_7RASvW1-qsa4XLg|file:///data/user/0/com.amazon.zordon/files/photo_booth_singl
e_shot2023-10-02_17-52-26-169.jpeg|2849322|photo_booth_single_shot2023-10-02_17-52-26-169.jpeg|0|D
ONE|0|0|0||1|05fec6d31b5ed3fae922fd92da90e3a1||AydH_yW8SVWt9Ei-YIWhTg|2849322|2849322|jpeg||||0||
2|photo_booth|5GxpwT_7RASvW1-qsa4XLg|file:///data/user/0/com.amazon.zordon/files/photo_booth_singl
e_shot2023-10-07_13-25-13-104.jpeg|3010907|photo_booth_single_shot2023-10-07_13-25-13-104.jpeg|0|D
ONE|0|0|0||1|22907c18e2bc9b38b19892cf3642d77c||MZSCkr7aSNGJneyNOGyV_Q|3010907|3010907|jpeg||||0|39
5|
3|photo_booth|5GxpwT_7RASvW1-qsa4XLg|file:///data/user/0/com.amazon.zordon/files/photo_booth_singl
e_shot2023-10-09_15-34-15-361.jpeg|3353396|photo_booth_single_shot2023-10-09_15-34-15-361.jpeg|0|D
ONE|0|0|0||1|d6a484656962aea314ae90cb7286ce14||AMuRAtPvTciCglQUfDBbMg|3353396|3353396|jpeg||||0|39
5|
4|photo_booth|5GxpwT_7RASvW1-qsa4XLg|file:///data/user/0/com.amazon.zordon/files/photo_booth_video
1696880081230.mp4|6047662|photo_booth_video1696880081230.mp4|0|DONE|0|0|0||1|79c220a31975e66d377e1
2e5b182b062||jtNsNiZBT5GWomjI5cxg7A|6047662|6047662|mp4||||0|395|
5|photo_booth|5GxpwT_7RASvW1-qsa4XLg|file:///data/user/0/com.amazon.zordon/files/photo_booth_singl
e_shot2023-10-09_19-45-07-628.jpeg|2935516|photo_booth_single_shot2023-10-09_19-45-07-628.jpeg|0|D
ONE|0|0|0||1|4c675e3d566f2df6b9f298acaab92c1d||fYRDY3pOTcmqyRfE9oUUBQ|2935516|2935516|jpeg||||0|39
5|
6|photo_booth|5GxpwT_7RASvW1-qsa4XLg|file:///data/user/0/com.amazon.zordon/files/photo_booth_singl
e_shot2023-10-17_20-29-28-381.jpeg|2717919|photo_booth_single_shot2023-10-17_20-29-28-381.jpeg|0|D
ONE|0|0|0||1|6823f8c4bb1b805f873a28a592a78beb||rr1nGSfQRnCXp_egEgIsMg|2717919|2717919|jpeg||||0|39
5|
7|photo_booth|5GxpwT_7RASvW1-qsa4XLg|file:///data/user/0/com.amazon.zordon/files/photo_booth_singl
e_shot2023-10-21_19-02-39-687.jpeg|2961976|photo_booth_single_shot2023-10-21_19-02-39-687.jpeg|0|D
ONE|0|0|0||1|f2acb3eecc567831c91e33682d2c6e06||wqNbnDllRnyGnmk_Q7eDkQ|2961976|2961976|jpeg||||0|39
5|
8|photo_booth|5GxpwT_7RASvW1-qsa4XLg|file:///data/user/0/com.amazon.zordon/files/photo_booth_video
1697929429134.mp4|13958448|photo_booth_video1697929429134.mp4|0|DONE|0|0|0||1|a4e2d785b8d73ba71a73
607cbccd9674||WHLskLUCRZWBW2hjBiLemQ|13958448|13958448|mp4||||0|395|
sqlite>
```

Upload Timestamp



```
sqlite> SELECT * FROM cloud_nodes;
_id|event_id|node_id|created_by|created_date|description|exclusively_trashed|is_root|is_shared|kin
d_id|modified_date|name|recursively_trashed|restricted|status_id|version|content_type|content_exte
nsion|content_md5|content_visual_digest|content_sort_date|content_date|content_size|content_versio
n|document_version|document_title|image_aperture_value|image_capture_mode|image_color_space|image_
date_time|image_date_time_digitized|image_date_time_original|image_exposure_mode|image_exposure_pr
ogram|image_exposure_time|image_flash|image_focal_length|image_gps_time_stamp|image_height|image_i
so_speed_ratings|image_location|image_make|image_metering_mode|image_model|image_orientation|image
_resolution_unit|image_sensing_method|image_sharpness|image_software|image_white_balance|image_wid
th|image_x_resolution|image_y_resolution|is_content|is_document|is_image|is_video|video_audio_bitr
ate|video_audio_channels|video_audio_channel_layout|video_audio_codec|video_audio_sample_rate|vide
o_bitrate|video_creation_date|video_duration|video_encoder|video_height|video_location|video_make|
video_model|video_rotate|video_title|video_overall_bitrate|video_codec|video_frame_rate|video_widt
h|favorite|hidden|metadata_version|has_share_parent|family_member_id
1|1|AydH_yW8SVWt9Ei-YIWhTg||1696283551557|||||1|1696283552715|photo_booth_single_shot2023-10-02_17
-52-26-169.jpeg||0|1|2|image/jpeg|jpg|05fec6d31b5ed3fae922fd92da90e3a1|64c52dcb95afb12fd1ad474ed91
95f27|1696283546015|1696269146015|2849322|1|||||sRGB|1696269146000|1696269146000|1696269146000|Aut
o|Not Defined|1/120|No Flash|3.8||2160|||Amazon|Center-weighted average|AEOAT|1|Pixels/Inch|||cust
om software|Auto|3456|||1|0|1|0|||||||||||||||||||||0|0|5|0|-1
2|2|MZSCkr7aSNGJneyNOGyV_Q||1696699520501|||||1|1696699521889|photo_booth_single_shot2023-10-07_13
-25-13-104.jpeg||0|1|2|image/jpeg|jpg|22907c18e2bc9b38b19892cf3642d77c|92e39c4a10b616f83c81eabefdf
51e30|1696699513094|1696685113094|3010907|1|||||sRGB|1696685113000|1696685113000|1696685113000|Aut
o|Not Defined|1/120|No Flash|3.8||2160|||Amazon|Center-weighted average|AEOAT|1|Pixels/Inch|||cust
om software|Auto|3456|||1|0|1|0|||||||||||||||||||||0|0|5|0|-1
3|3|AMuRAtPvTciCglQUfDBbMg||1696880061012|||||1|1696880062078|photo_booth_single_shot2023-10-09_15
-34-15-361.jpeg||0|1|2|image/jpeg|jpg|d6a484656962aea314ae90cb7286ce14|9b952465994204e06f43c178508
872e9|1696880055034|1696865655034|3353396|1|||||sRGB|1696865655000|1696865655000|1696865655000|Aut
o|Not Defined|1/120|No Flash|3.8||2160|||Amazon|Center-weighted average|AEOAT|1|Pixels/Inch|||cust
om software|Auto|3456|||1|0|1|0|||||||||||||||||||||0|0|5|0|-1
4|4|jtNsNiZBT5GWomjI5cxg7A||1696880095819|||||1|1696880097618|photo_booth_video1696880081230.mp4||
0|1|3|video/mp4|mp4|79c220a31975e66d377e12e5b182b062||1696880095819||6047662|1|||||||||||||||||||
||||||||||1|0|0|0|||||||||||||||||||||0|0|5|0|-1
5|5|fYRDY3pOTcmqyRfE9oUUBQ||1696895111692|||||1|1696895113183|photo_booth_single_shot2023-10-09_19
-45-07-628.jpeg||0|1|2|image/jpeg|jpg|4c675e3d566f2df6b9f298acaab92c1d|cddae4f76d9581d069ab342d512
5ed2b|1696895107061|1696880707061|2935516|1|||||sRGB|1696880707000|1696880707000|1696880707000|Aut
o|Not Defined|1/120|No Flash|3.8||2160|||Amazon|Center-weighted average|AEOAT|1|Pixels/Inch|||cust
om software|Auto|3456|||1|0|1|0|||||||||||||||||||||0|0|5|0|-1
6|6|rr1nGSfQRnCXp_egEgIsMg||1697588974960|||||1|1697588975964|photo_booth_single_shot2023-10-17_20
-29-28-381.jpeg||0|1|2|image/jpeg|jpg|6823f8c4bb1b805f873a28a592a78beb|2fe249c93438cd089de50bff10c
```

Convert epoch time to UTC



```
7f7ef|1697929359067|1697914959067|2961976|1|||||sRGB|1697914959000|1697914959000|1697914959000|Aut
o|Not Defined|1/24|No Flash|3.8||2160|||Amazon|Center-weighted average|AEOAT|1|Pixels/Inch|||custo
m software|Auto|3456|||1|0|1|0|||||||||||||||||||||0|0|5|0|-1
8|8|WHLskLUCRZWBW2hjBiLemQ||1697929464021|||||1|1697929467776|photo_booth_video1697929429134.mp4||
0|1|3|video/mp4|mp4|a4e2d785b8d73ba71a73607cbccd9674||1697929464021||13958448|1|||||||||||||||||||
||||||||||1|0|0|0|||||||||||||||||||||0|0|5|0|-1
sqlite> .exit

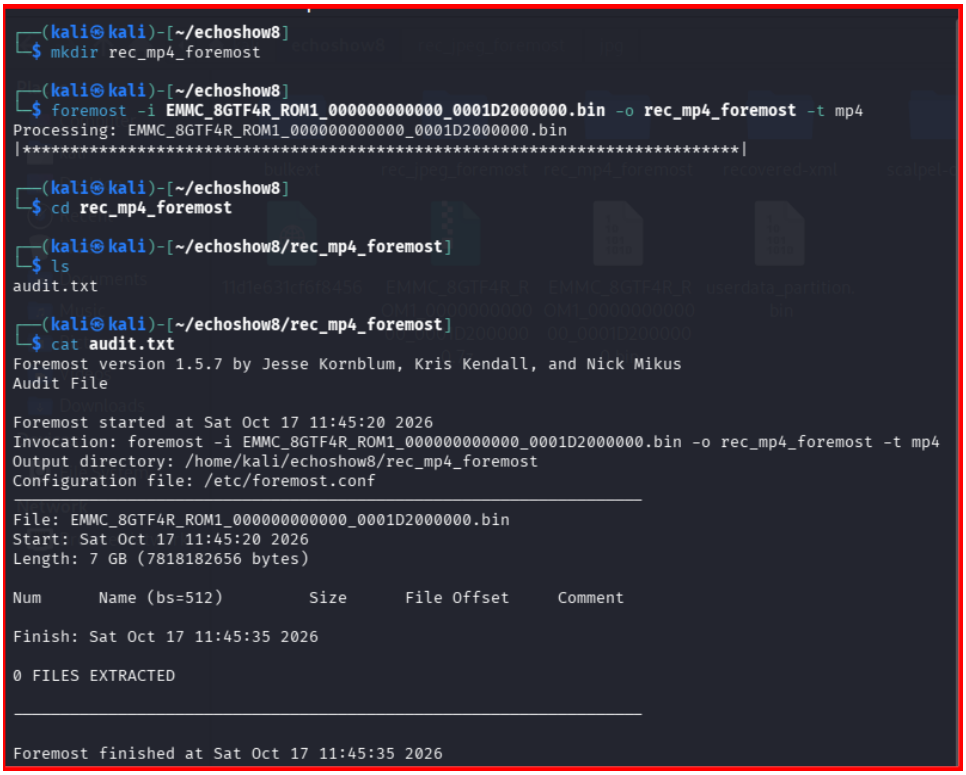
┌──(root㉿kali)-[/mnt/echoshow8-2/data/com.amazon.zordon/databases]
└─# date -u -d @1697929372
Sat Oct 21 11:02:52 PM UTC 2023

┌──(root㉿kali)-[/mnt/echoshow8-2/data/com.amazon.zordon/databases]
└─# date -u -d @1697929464
Sat Oct 21 11:04:24 PM UTC 2023
```

Videos Saved on Device

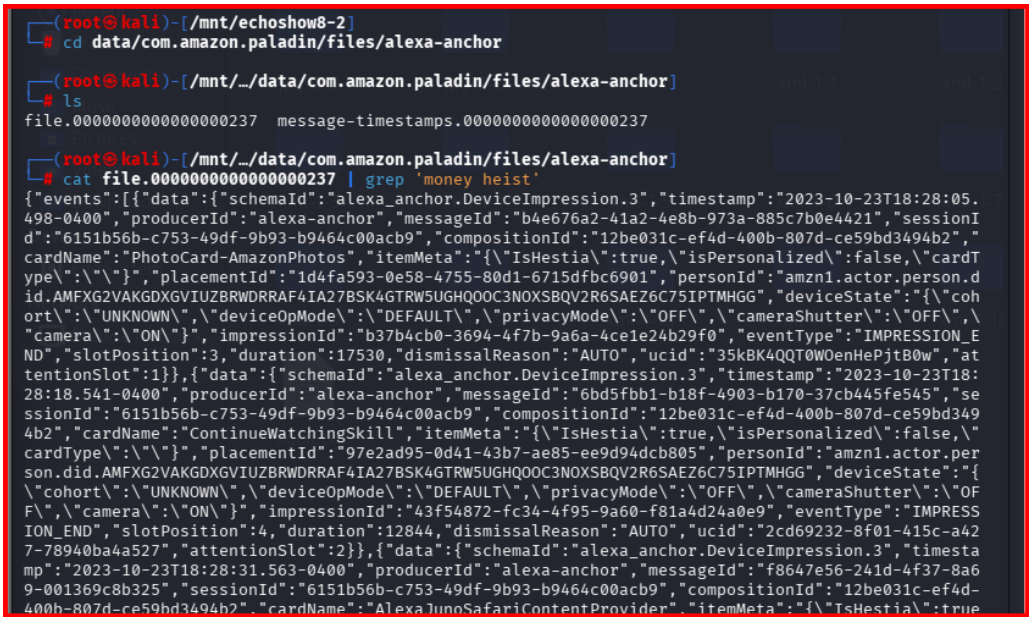
Search Videos by Carving mp4 files

Create a directory and run foremost



NO VIDEOS WERE FOUND EVEN IN UNALLOCATED SPACE.

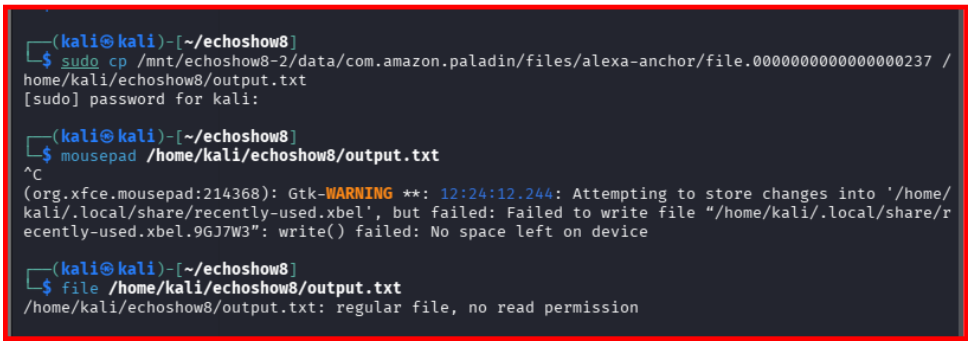
MUSIC (SONGS) PLAYED ON DEVICE (ECHO SHOW LOG FILE)




7-78940ba4a527","attentionSlot":2}},{"data":{"schemaId":"alexa_anchor.DeviceImpression.3","timesta
mp":"2023-10-23T18:28:31.563-0400","producerId":"alexa-anchor","messageId":"f8647e56-241d-4f37-8a6
9-001369c8b325","sessionId":"6151b56b-c753-49df-9b93-b9464c00acb9","compositionId":"12be031c-ef4d-
400b-807d-ce59bd3494b2","cardName":"AlexaJunoSafariContentProvider","itemMeta":"{\"IsHestia\":true
,\"isPersonalized\":false,\"cardType\":\"\"}","placementId":"8e05df21-2e2a-4fb8-bd60-f847258d7f20"
,"personId":"amzn1.actor.person.did.AMFXG2VAKGDXGVIUZBRWDRRAF4IA27BSK4GTRW5UGHQOOC3NOXSBQV2R6SAEZ6
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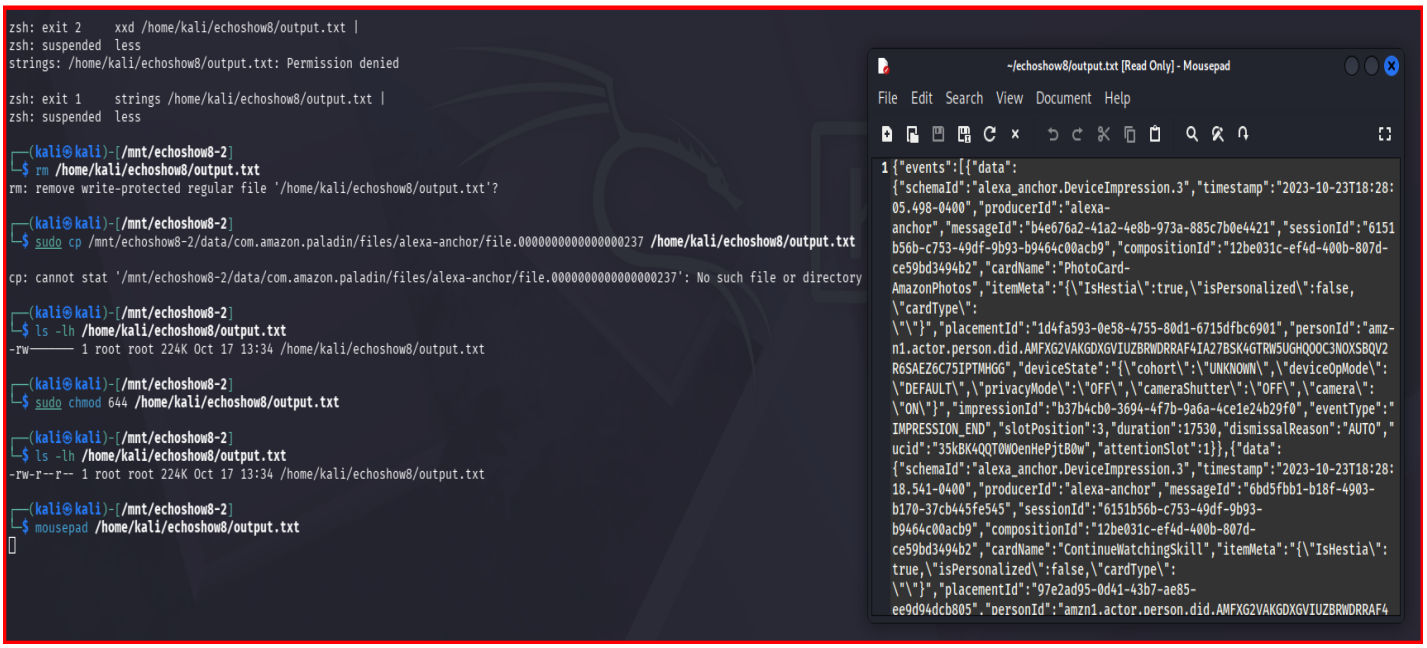

rdName":"PhotoCard-AmazonPhotos","itemMeta":"{\"IsHestia\":true,\"isPersonalized\":false,\"cardTyp
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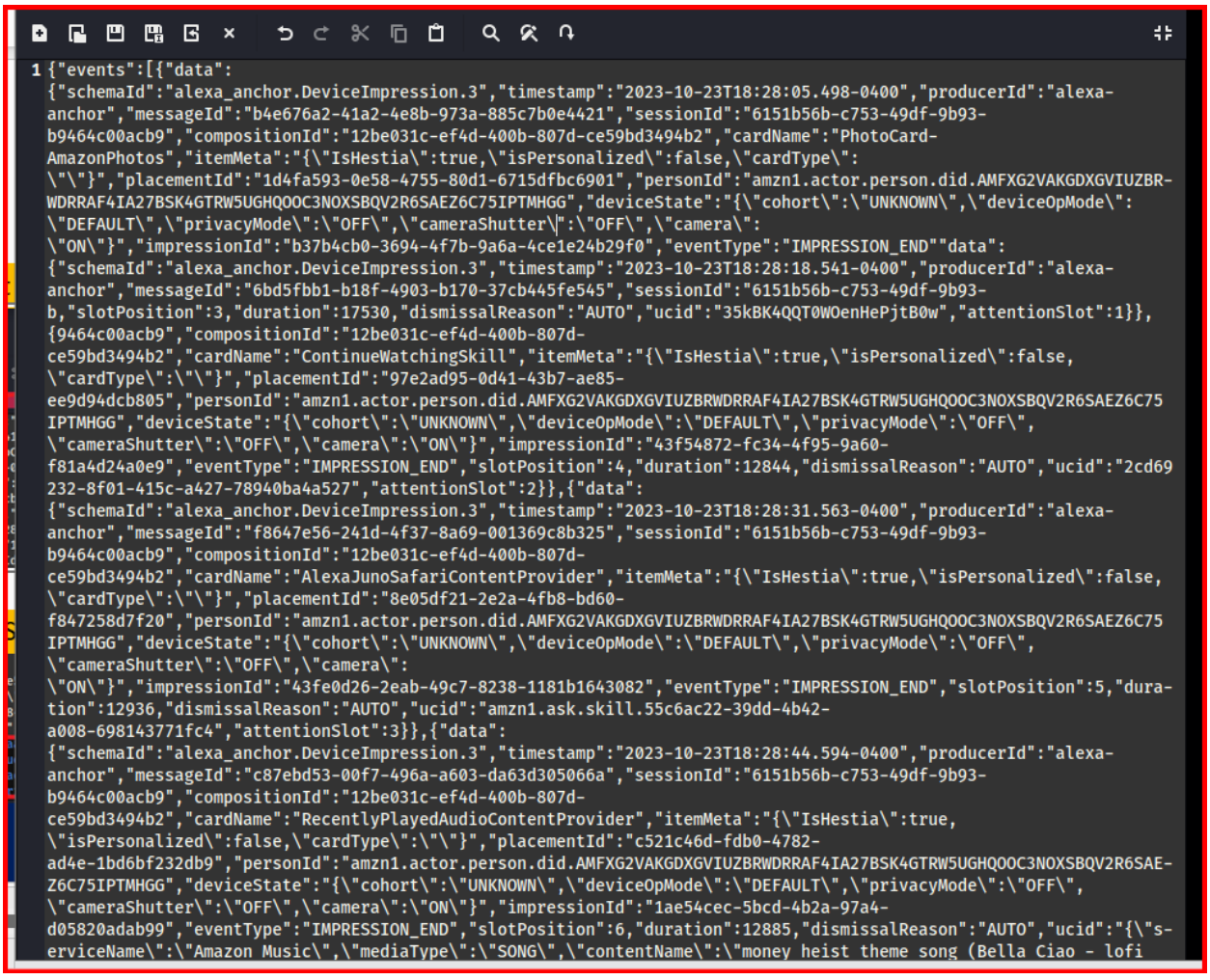
┌──(root㉿kali)-[/mnt/…/data/com.amazon.paladin/files/alexa-anchor]
└─#


┌──(kali㉿kali)-[~/echoshow8]
└─$ sudo cp /mnt/echoshow8-2/data/com.amazon.paladin/files/alexa-anchor/file.000000000000000237 /
home/kali/echoshow8/output.txt
[sudo] password for kali:

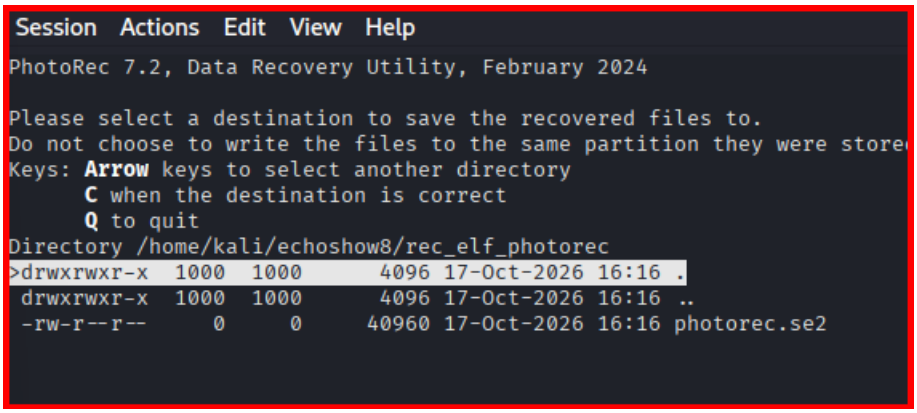
┌──(kali㉿kali)-[~/echoshow8]
└─$ mousepad /home/kali/echoshow8/output.txt
^C
(org.xfce.mousepad:214368): Gtk-WARNING **: 12:24:12.244: Attempting to store changes into '/home/
kali/.local/share/recently-used.xbel', but failed: Failed to write file “/home/kali/.local/share/r
ecently-used.xbel.9GJ7W3”: write() failed: No space left on device

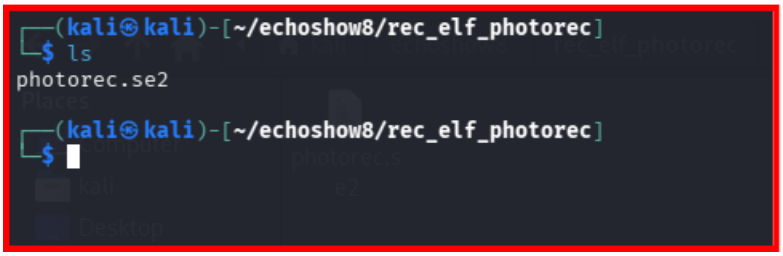
┌──(kali㉿kali)-[~/echoshow8]
└─$ file /home/kali/echoshow8/output.txt
/home/kali/echoshow8/output.txt: regular file, no read permission

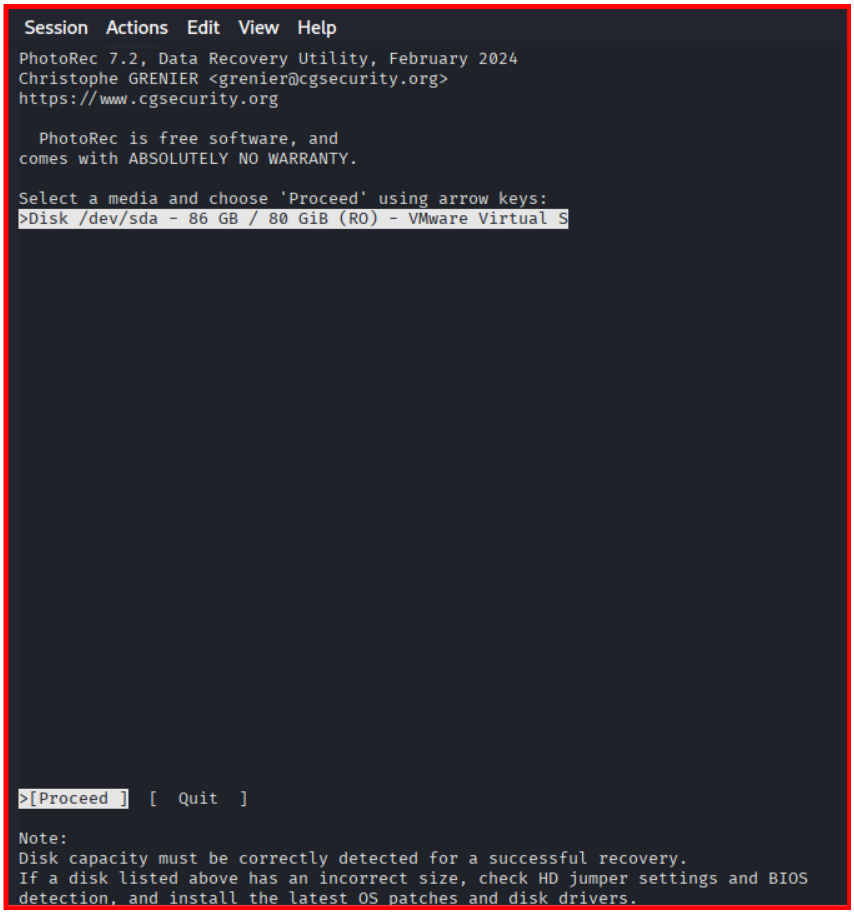

zsh: exit 2 xxd /home/kali/echoshow8/output.txt |
zsh: suspended less
strings: /home/kali/echoshow8/output.txt: Permission denied
zsh: exit 1 strings /home/kali/echoshow8/output.txt |
zsh: suspended less
(kali㉿kali)-[/mnt/echoshow8-2]
$ rm /home/kali/echoshow8/output.txt
rm: remove write-protected regular file '/home/kali/echoshow8/output.txt'?
(kali㉿kali)-[/mnt/echoshow8-2]
$ sudo cp /mnt/echoshow8-2/data/com.amazon.paladin/files/alexa-anchor/file.000000000000000237 /home/kali/echoshow8/output.txt
cp: cannot stat '/mnt/echoshow8-2/data/com.amazon.paladin/files/alexa-anchor/file.000000000000000237': No such file or directory
(kali㉿kali)-[/mnt/echoshow8-2]
$ ls -lh /home/kali/echoshow8/output.txt
-rw——— 1 root root 224K Oct 17 13:34 /home/kali/echoshow8/output.txt
(kali㉿kali)-[/mnt/echoshow8-2]
$ sudo chmod 644 /home/kali/echoshow8/output.txt
(kali㉿kali)-[/mnt/echoshow8-2]
$ ls -lh /home/kali/echoshow8/output.txt
-rw-r--r-- 1 root root 224K Oct 17 13:34 /home/kali/echoshow8/output.txt
(kali㉿kali)-[/mnt/echoshow8-2]
$ mousepad /home/kali/echoshow8/output.txt
~/echoshow8/output.txt [Read Only] - Mousepad
File Edit Search View Document Help
1 {"events":[{"data":
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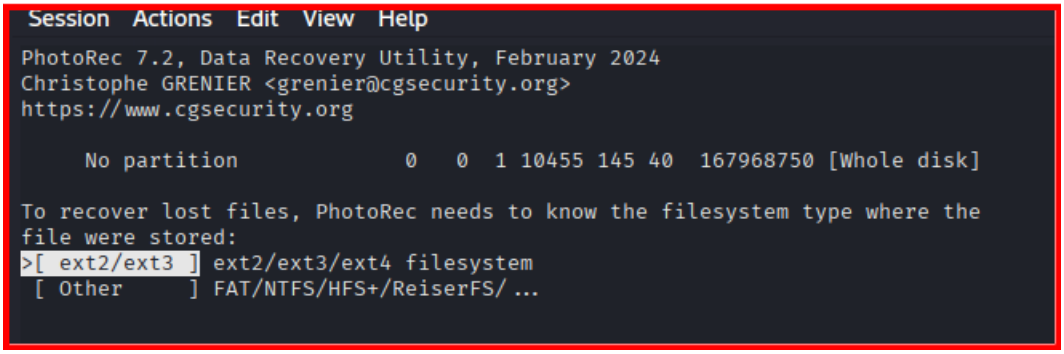

1 {"events":[{"data":
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{9464c00acb9","compositionId":"12be031c-ef4d-400b-807d-ce59bd3494b2","cardName":"ContinueWatchingSkill","itemMeta":"{\"IsHestia\":true,\"isPersonalized\":false,\"cardType\":\"\"}","placementId":"97e2ad95-0d41-43b7-ae85-ee9d94dcb805","personId":"amzn1.actor.person.did.AMFXG2VAKGDXGVIUZBRWDRRAF4IA27BSK4GTRW5UGHQOOC3NOXSBQV2R6SAEZ6C75IPTMHGG","deviceState":"{\"cohort\":\"UNKNOWN\",\"deviceOpMode\":\"DEFAULT\",\"privacyMode\":\"OFF\",\"cameraShutter\":\"OFF\",\"camera\":\"ON\"}","impressionId":"43f54872-fc34-4f95-9a60-f81a4d24a0e9","eventType":"IMPRESSION_END","slotPosition":4,"duration":12844,"dismissalReason":"AUTO","ucid":"2cd69232-8f01-415c-a427-78940ba4a527","attentionSlot":2}},{"data":
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Carving ‘elf’ Files Using Photorec









Conclusion

This conclusion answers the case question using only the Echo Show 8 training evidence examined in this report. The examination used the eMMC image *EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin*. The examiner attached the image as a read-only loop device. The system partition (p21) and userdata partition (p23) were then mounted for analysis, and the userdata partition was mounted with the *noload* option so that the ext4 journal would not be replayed. The image hash matched the value in the supplied instruction file. The examination can establish the device's identity, operating system, primary user, account-linked identifiers, Wi-Fi and Bluetooth configuration, web and search activity, a recovered phone call record and several multimedia artefacts. It cannot establish who physically operated the device, and it does not support any finding about whether an offence was committed. Values shown as [insert] appear only in screenshots or tool output. They must be copied exactly from those sources before the report is finalised.

Limitations and Unknowns

- **Evidence package record:** Section 2 does not yet record the archive name, archive size, image file name list, supplied manifest or the hash values themselves. These details must be added so that the chain of evidence can be audited.
- **Derived working copies:** The userdata partition was extracted with dd (skip=7929856, count=7339999 sectors of 512 bytes). This extract should be hashed, and its offsets should be checked against the mmls output, so that the carved results can be reproduced.

- **Carved artefacts:** Files recovered with scalpel, foremost and PhotoRec have no file system path. Their original names and timestamps may also be lost, so the time and origin of the carved camera image and the carved call record depend on the content of each file. scalpel produced more than 27 GB of output, much of it duplicated, so the results should be deduplicated by hash.
- **Uploaded media:** The records show that 6 photos and 2 videos were uploaded, but no MP4 files were recovered. The content of the uploaded videos therefore cannot be examined from the device. Confirming them would require cloud data, which is outside the scope of this examination.
- **Attribution:** The account email address, physical address, user profile and Wi-Fi details link the device to an account and a location. They do not identify who used the device at any given time. Calls, searches and media activity are attributed only to the device or the device user.
- **Time:** The time zone and clock accuracy of the device have not been independently verified. All converted times should state the time zone used.
- **Unknown:** The examination did not establish the content of the call, whether anyone answered it, or how long it lasted beyond what the carved record shows. It also did not establish any activity that only the cloud would hold.

Overall Conclusion

A careful, read-only examination of the Echo Show 8 eMMC and userdata evidence can reliably establish several facts. These are the device's identity and Fire OS configuration, a single primary user (ID 0), and identifiers linked to the Amazon account. The examination also establishes the device's Wi-Fi and Bluetooth connections, its web and search activity, and a recovered call record for the contact 'hibna'. Finally, it establishes the multimedia activity: a carved camera image, records of 6 photos and 2 videos being uploaded, and video and music playback. File system analysis alone would have missed important evidence. The phone logs and the camera image were recovered only by carving unallocated space, which shows that carving is necessary when examining this type of device. The findings describe what the device recorded. They do not identify the person operating it or show any intent. The chronology can be presented as defensible once each [insert] value has been copied from its source output, its time zone stated, and the evidence package details recorded in Section 2.